

Structured Phase Transitions on a Finite Interval

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1. Introduction

VAN DER WAALS, in his classic paper [1893],¹ gave arguments in support of a compressible fluid whose free energy at constant temperature depends not only on the density, but also on the *density gradient*. CAHN & HILLIARD [1958], apparently unaware of VAN DER WAALS' paper, rederived VAN DER WAALS' theory and, using this theory, obtained several important results concerning the interfacial energy between phases. Since then gradient theories have been used to analyze phase transitions, spinodal decomposition, and other physical phenomena.²

In VAN DER WAALS' theory, restricted to one space variable x , the energy of fluid occupying a vessel of unit cross section, extending from $x = -L$ to $x = L$, is

$$\int_{-L}^L [W(u(x)) + \varepsilon^2 u'(x)^2] dx. \quad (1.1)$$

Here $u(x)$ is the fluid density, $W(u)$ is the free energy per unit volume when the density is constant, and $\varepsilon > 0$ is a small parameter. If the total mass in the container is M , then we have the additional constraint

$$\int_{-L}^L u(x) dx = M. \quad (1.2)$$

VAN DER WAALS, following GIBBS, believed that the stable configurations of the fluid are those which minimize (1.1) subject to (1.2). In modern terminology, this suggests the problem:

($\mathcal{P}_\varepsilon$) minimize (1.1) over all $u \in H^1(-L, L)$, $u > 0$, which satisfy the constraint (1.2).

¹ Translated by ROWLINSON [1979].

² Cf. ROWLINSON [1979] and DAVIS & SCRIVEN [1982] for selected references.

Here $H^1(-L, L)$ is the usual Sobolev space of square-integrable functions possessing square-integrable first derivatives. Note that both (1.1) and (1.2) are invariant under *reversal*; that is, under the transformation $u(x) \rightarrow u(-x)$.

For $W \in C^2(0, \infty)$ the direct method of the calculus of variations and elementary regularity theory lead to the conclusion¹ that Problem $\mathcal{P}_\varepsilon$ possesses a (not necessarily unique) solution in $C^2[-L, L]$, so existence is not at issue here. The goal instead is to identify the minimizers of $\mathcal{P}_\varepsilon$ when the "chemical potential" $W'(u)$ has the form illustrated in Figure 1, a form motivated by the original potential of van der Waals.

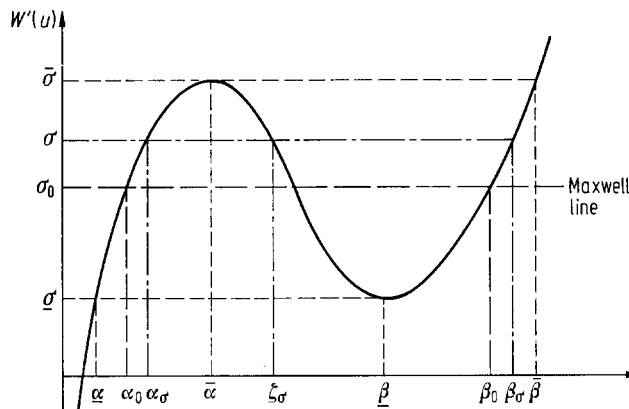


Fig. 1

Consider first the problem with $\varepsilon = 0$, for which (1.1) has the form

$$\int_{-L}^L W(u(x)) \, dx. \quad (1.3)$$

This problem may be stated as follows:

$(\mathcal{P}_0)$ minimize (1.3)—subject to (1.2)—over all $u > 0$ with $u, W(u) \in L^1(-L, L)$.

$\mathcal{P}_0$ is easily solved with the aid of the auxiliary functional

$$\int_{-L}^L [W(u(x)) - \sigma u(x)] \, dx$$

in which σ (= constant) is a Lagrange multiplier. For a minimum to exist the Euler-Lagrange equation and Weierstrass-Erdmann corner conditions must be satisfied:

$$\begin{aligned} W'(u) &= \sigma \text{ at points of continuity of } u, \text{ while} \\ W(u) - \sigma u &\text{ is continuous across jumps in } u. \end{aligned} \quad (1.4)$$

¹ Cf. MORREY [1966], Theorems 1.9.1 and 10.1. Actually, the constraint $u > 0$ causes some difficulty in the case $W'(0^+) = -\infty$, but a test-function argument can be used to show that if u is a minimizer, then u is necessarily bounded away from zero.

Inspection of (1.4) shows that solutions are either constant (*single phase*) or piecewise constant (*two phase*), and in the latter case have the form

$$u(x) = \begin{cases} \alpha_0, & x \in S_1 \\ \beta_0, & x \in S_2, \end{cases} \quad (1.5)$$

where S_1 and S_2 are disjoint measurable sets whose union is $[-L, L]$, while α_0 and β_0 are defined by the *Maxwell conditions* (cf. Figure 1)

$$\begin{aligned} W(\beta_0) - W(\alpha_0) &= \sigma_0(\beta_0 - \alpha_0), \\ \sigma_0 &= W'(\alpha_0) = W'(\beta_0). \end{aligned} \quad (1.6)$$

Further, letting

$$l_i = \text{measure}(S_i),$$

(1.2) yields

$$l_1 = \frac{2L(\beta_0 - r)}{\beta_0 - \alpha_0}, \quad l_2 = \frac{2L(r - \alpha_0)}{\beta_0 - \alpha_0} \quad (1.7)$$

with

$$r = \frac{M}{2L}$$

the *average density*. Since $l_i > 0$, a necessary condition for the existence of a two-phase solution is that

$$\alpha_0 < r < \beta_0. \quad (1.8)$$

When (1.8) is satisfied, any $u(x)$ of the form (1.5) with l_i given by (1.7) is a global minimizer for Problem $\mathcal{P}_0$.

If $r \leq \alpha_0$ or $r \geq \beta_0$ a two-phase solution of $\mathcal{P}_0$ is not possible; the minimizer is simply the single-phase solution

$$u(x) \equiv r.$$

As the above discussion shows, for $\alpha_0 < r < \beta_0$ there are infinitely many solutions to $\mathcal{P}_0$. Here we shall attempt to answer the question:

Are any of the two-phase solutions—in some physical sense—preferred?

Indeed, the theory with $\varepsilon = 0$ allows the formation of interfaces (jumps in density) without a concomitant increase in energy. One might expect that in a theory which includes interfacial energy, the two-phase solutions with least energy would be the *single-interface* solutions

$$u_0(x) = \begin{cases} \alpha_0, & -L \leq x \leq -L + l_1 \\ \beta_0, & -L + l_1 < x \leq L \end{cases} \quad \text{and } u_0(-x). \quad (1.9)$$

We here show that this expectation is, in fact, justified.

Consider the theory with $\varepsilon > 0$. While this theory does not allow for jumps in density, it does allow u to suffer rapid changes over small intervals, and such changes are penalized in energy by the term $\varepsilon^2 u'(x)^2$ in (1.1). Thus the theory with $\varepsilon > 0$ has associated with it a natural interfacial energy (cf. CAHN &

HILLIARD [1958]). Our main result is that for $r \in (\alpha_0, \beta_0)$:

(i) when $\varepsilon > 0$ is small enough, Problem $\mathcal{P}_\varepsilon$ has a unique (modulo reversal) global minimizer $u_\varepsilon(x)$;

(ii) $u_\varepsilon(x)$ is strictly monotone;

(iii) as $\varepsilon \rightarrow 0$, $u_\varepsilon(x)$ (or its reversal) approaches the single-interface solution $u_0(x)$ of (1.9).

The single-interface solutions (1.9) are therefore preferred in the sense that they represent limits, as $\varepsilon \rightarrow 0$, of solutions within the VAN DER WAALS' theory.

Our proof of (i)–(iii) is based on a systematic study of the associated system, $\mathcal{S}_\varepsilon$, consisting of the Euler-Lagrange equation, the natural boundary conditions, and the constraint.¹ One step in the proof is to show that monotonic solutions of $\mathcal{S}_\varepsilon$ have lower energy than nonmonotonic solutions. In fact, we show that:

(iv) nonmonotonic solutions of $\mathcal{S}_\varepsilon$ are unstable in the sense that they cannot be even local minimizers for $\mathcal{P}_\varepsilon$.

The theory and results discussed thus far have possible application to the study of phase transformations in elastic solids. Consider a one-dimensional elastic filament which occupies the interval $[-L, L]$ in a fixed reference configuration. For each material point $x \in [-L, L]$, let $f(x)$ denote the position of x in the deformed configuration, and let

$$u(x) = f'(x)$$

designate the “strain”.

Using as a basis VAN DER WAALS' argument, CARR, GURTIN, & SLEMROD [1983] considered a stored-energy function of the form²

$$W(u) + \varepsilon^2(u')^2 \tag{1.10}$$

with $W(u)$ —the stress in the filament when u is constant—of the type illustrated in Figure 1. The functional (1.1) then represents the total strain energy, while the constraint (1.2) is the requirement that the bar have length M in the deformed configuration. Thus for materials defined by (1.10), Problem $\mathcal{P}_\varepsilon$ corresponds to the hard-loading device of ERICKSEN [1975], in which the displacement is specified at the ends of the filament³. The soft device, in which the force is specified, has been studied in detail by CARR, GURTIN, & SLEMROD [1983].

¹ Cf. NOVICK-COHEN & SEGEL [1984] and NOVICK-COHEN [1983], who obtain solutions of $\mathcal{S}_\varepsilon$ in terms of Jacobi elliptic functions for $W(u)$ quartic in u . To our knowledge these are the first authors to consider $\mathcal{S}_\varepsilon$ with L finite. There is a large and rapidly growing literature—which we make no attempt to discuss—on associated problems with $L = \infty$. In this connection cf., e.g., CAHN & HILLIARD [1958], LANGER [1971], JACKIW [1977], DAVIS & SCRIVEN [1982], AIFANTIS & SERRIN [1983], and COLEMAN [1983].

² A general theory of elastic materials with stored energy dependent on strain gradients was proposed by TOUPIN [1962]; ERICKSEN [1975] has suggested a nonconvex W as a model for solids undergoing phase transitions.

³ Cf. ANTMAN [1972, 1973] and ANTMAN & CARBONE [1977], who establish—for a class of nonlinear hyperelastic rods in tension—the existence of solutions of the type that describe necking. Within our theory a neck would be a solution with two transitions and as such would be unstable (cf. (iv) above).

Our results also have application to the theory of binary mixtures as developed by CAHN & HILLIARD [1958]. There u represents concentration and is restricted to the interval $0 \leq u \leq 1$. (This additional restriction on u causes no difficulties.)

For convenience, we shall assume the length scale chosen so that

$$L = 1.$$

2. The problem

The problem under consideration may be stated precisely as follows:

($\mathcal{P}_{er}$) minimize

$$\int_{-1}^1 [W(u(x)) + \varepsilon^2 u'(x)^2] dx \quad (2.1)$$

over all $u \in H^1(-1, 1)$, $u > 0$, consistent with the constraint

$$\int_{-1}^1 u(x) dx = 2r. \quad (2.2)$$

We assume that W' has the form shown in Figure 1; more precisely, we suppose that

- (i) $W: (0, \infty) \rightarrow \mathbb{R}$ is C^5 ;
- (ii) $W'' > 0$ on $(0, \bar{\alpha}) \cup (\beta, \infty)$, $W'' < 0$ on $(\bar{\alpha}, \beta)$;
- (iii) $W'(0) < W'(\beta)$, $W'(\infty) > W'(\bar{\alpha})$.

The classical theory of the calculus of variations asserts that a smooth minimizer exists, and that any such minimizer must satisfy the corresponding Euler-Lagrange equation and natural boundary conditions. If we write

$$x = \varepsilon t, \quad z(t) = u(\varepsilon t),$$

these and the constraint (2.2) take the form

$$\mathcal{P}_{er} \begin{cases} 2\ddot{z} = W'(z) - \sigma, & (2.3) \\ \dot{z}(\pm \varepsilon^{-1}) = 0, & (2.4) \\ \int_{-\varepsilon^{-1}}^{\varepsilon^{-1}} z(t) dt = 2r\varepsilon^{-1}, & (2.5) \end{cases}$$

where σ (= constant) is a Lagrange multiplier.

Equation (2.3) has the first integral

$$\dot{z}^2 = \Phi_\sigma(z) - b \quad (2.6)$$

with b a constant and

$$\Phi_\sigma(z) = W(z) - \sigma z \quad (2.7)$$

the associated Gibbs function. Note that, by (2.4),

$$\Phi_\sigma(z_1) = \Phi_\sigma(z_2) = b, \quad (2.8)$$

where

$$z_1 = z(-\varepsilon^{-1}), \quad z_2 = z(\varepsilon^{-1}) \quad (2.9)$$

are the boundary-values of $z(t)$.

Proposition 2.1 (Properties of the Gibbs function). *The mapping $(z, \sigma) \mapsto \Phi_\sigma(z)$ is C^5 on $(0, \infty) \times \mathbb{R}$. Further, for each $\sigma \in (\underline{\sigma}, \bar{\sigma})$:*

(i) Φ_σ has exactly three critical points $\alpha_\sigma < \xi_\sigma < \beta_\sigma$:

$$\Phi'_\sigma(\alpha_\sigma) = \Phi'_\sigma(\xi_\sigma) = \Phi'_\sigma(\beta_\sigma) = 0; \quad (2.10)$$

(ii) Φ_σ is strictly decreasing on $(0, \alpha_\sigma) \cup (\xi_\sigma, \beta_\sigma)$, strictly increasing on $(\alpha_\sigma, \xi_\sigma) \cup (\beta_\sigma, \infty)$;

(iii) $\Phi_\sigma(\beta_\sigma) > \Phi_\sigma(\alpha_\sigma)$ for $\sigma < \sigma_0$, $\Phi_\sigma(\alpha_\sigma) > \Phi_\sigma(\beta_\sigma)$ for $\sigma > \sigma_0$,

$$\Phi_{\sigma_0}(\alpha_0) = \Phi_{\sigma_0}(\beta_0). \quad (2.11)$$

Proof. The critical points ξ_σ of Φ_σ are obviously solutions of

$$W'(\xi_\sigma) = \sigma, \quad (2.12)$$

and (i) and (ii) are immediate consequences of the properties of W' (cf. Figure 1). Further, (2.7) and (2.12) imply that

$$\frac{d}{d\sigma} \Phi_\sigma(\xi_\sigma) = -\xi_\sigma \quad (2.13)$$

and therefore

$$\frac{d}{d\sigma} [\Phi_\sigma(\alpha_\sigma) - \Phi_\sigma(\beta_\sigma)] = \beta_\sigma - \alpha_\sigma > 0. \quad (2.14)$$

On the other hand, (1.6) implies (2.11), and this with (2.14) yields the inequalities in (iii). $\square$

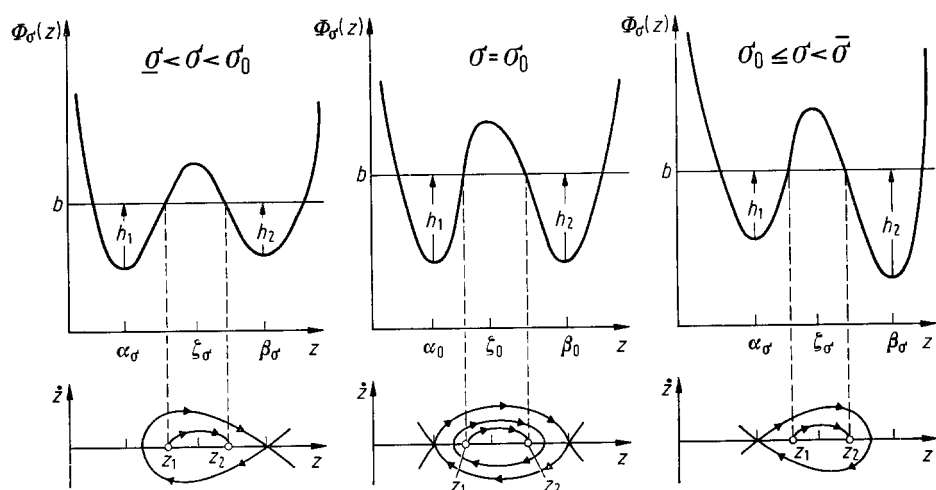


Fig. 2. The Gibbs function and the phase portraits, the points $z_i = z_i(\Delta)$, and the parameters h_i

Because of the boundary condition (2.4), our problem reduces to finding a trajectory-segment which begins and ends on the axis $\dot{z} = 0$ and which has duration $2\varepsilon^{-1}$. Thus nonconstant solutions of $\mathcal{S}_{\varepsilon}$ correspond to periodic solutions and are hence possible only for $\sigma \in (\underline{\sigma}, \bar{\sigma})$. We therefore restrict our attention to this interval. Further, it is clear from Figure 2 that for such σ the constant b will have the range

$$\begin{aligned}\Phi_{\sigma}(\beta_{\sigma}) < b < \Phi_{\sigma}(\zeta_{\sigma}), & \quad \sigma \leq \sigma_0, \\ \Phi_{\sigma}(\alpha_{\sigma}) < b < \Phi_{\sigma}(\zeta_{\sigma}), & \quad \sigma \geq \sigma_0.\end{aligned}\tag{2.15}$$

A pair

$$\Delta = (\sigma, b)$$

with $\underline{\sigma} < \sigma < \bar{\sigma}$ and b consistent with (2.15) will be called *admissible*.

It is convenient to rewrite (2.6) and (2.8) in the form

$$\begin{aligned}\dot{z}^2 &= f_{\Delta}(z), \\ f_{\Delta}(z_1) &= f_{\Delta}(z_2) = 0,\end{aligned}\tag{2.16}$$

where

$$f_{\Delta}(z) = \Phi_{\sigma}(z) - b.\tag{2.17}$$

Our next result allows us to consider the boundary values z_i as functions of Δ .

Proposition 2.2. *For each admissible Δ there is exactly one point $z_1(\Delta) \in (\sigma_{\sigma}, \zeta_{\sigma})$ and exactly one point $z_2(\Delta) \in (\zeta_{\sigma}, \beta_{\sigma})$ such that*

$$f_{\Delta}(z_i(\Delta)) = 0.\tag{2.18}$$

Proof. Let g_{σ} denote the restriction of Φ_{σ} to $[\alpha_{\sigma}, \zeta_{\sigma}]$. Then, since Φ_{σ} is strictly increasing on $[\alpha_{\sigma}, \zeta_{\sigma}]$, (cf. Proposition 2.1) and since b belongs to the range of g_{σ} when Δ is admissible, (2.18) has the unique solution

$$z_1(\Delta) = g_{\sigma}^{-1}(b)\tag{2.19}$$

in $(\alpha_{\sigma}, \zeta_{\sigma})$. A similar argument applies to $z_2(\Delta)$. $\square$

Let us agree to use the term *simple solution* for a *nonconstant* solution $z(t)$ of $\mathcal{S}_{\varepsilon}$ with $\dot{z}(t) \geq 0$. Let $z(t)$ be a simple solution and let $\Delta = (\sigma, b)$ be the *corresponding admissible pair* (computed using (2.3) and (2.6)). The requirement that $z(t)$ have duration $2\varepsilon^{-1}$ and the constraint (2.5) yield additional restrictions on Δ , which we now derive.

Since $\dot{z}(t) > 0$ on $(-\varepsilon^{-1}, \varepsilon^{-1})$, we may use (2.9) and (2.16)₁ to convert the integral

$$\int_{-\varepsilon^{-1}}^{\varepsilon^{-1}} z(t)^n dt$$

into one of the form

$$I_n(\Delta) = \int_{z_1(\Delta)}^{z_2(\Delta)} z^n f_{\Delta}(z)^{-\frac{1}{2}} dz.\tag{2.20}$$

Thus, since $n = 0$ yields the duration of $z(t)$, while $n = 1$ leads to the constraint integral in (2.5),

$$\begin{aligned} I_0(\Delta) &= 2\varepsilon^{-1}, \\ I_1(\Delta) &= 2r\varepsilon^{-1}. \end{aligned} \quad (2.21)$$

Theorem 2.1. *There is a one-to-one correspondence between simple solutions $z(t)$ and admissible pairs Δ that satisfy (2.21): given any simple solution, the corresponding admissible pair Δ satisfies (2.21); given any admissible pair Δ that satisfies (2.21), there is exactly one simple solution $z(t)$ to which Δ corresponds.*

Proof. We have only to establish the converse assertion. Let $\Delta = (\sigma, b)$ be admissible and satisfy (2.21). Let $z_i = z_i(\Delta)$ be the points established in Proposition 2.2. Then it is clear from the phase portrait associated with this value of σ that there is exactly one trajectory $z(t)$ which begins at z_1 with $\dot{z} = 0$, ends at z_2 with $\dot{z} = 0$, and has $\dot{z}(t) \geq 0$. By (2.21) this trajectory had duration $2\varepsilon^{-1}$ and satisfies the constraint (2.5). $\square$

3. Statement of the main result. Sketch of the proof

Let

$$\Delta_0 = (\sigma_0, b_0), \quad b_0 = \Phi_{\sigma_0}(\alpha_0) = \Phi_{\sigma_0}(\beta_0). \quad (3.1)$$

We will refer to Δ_0 as the *Maxwell point*; since

$$z_1(\Delta_0) = \alpha_0, \quad z_2(\Delta_0) = \beta_0, \quad (3.2)$$

Δ_0 corresponds to the heteroclinic orbit from α_0 to β_0 .

We now state our main result; its proof will require much of the remainder of the paper. For convenience, we write

$$\mathcal{R}_\delta = [\alpha_0 + \delta, \beta_0 - \delta]. \quad (3.3)$$

Theorem 3.1 (The Maxwell solution). *Given $\delta > 0$, there are an $\varepsilon_\delta > 0$ and a neighborhood $\mathcal{N}_\delta$ of Δ_0 such that: for $\varepsilon \in (0, \varepsilon_\delta)$ and $r \in \mathcal{R}_\delta$, Problem $\mathcal{P}_{\varepsilon r}$ has exactly one simple solution with corresponding admissible pair $\Delta(\varepsilon, r) \in \mathcal{N}_\delta$. Further, there is a constant $C = C_\delta > 0$ such that*

$$\Delta(\varepsilon, r) - \Delta_0, \quad z_1(\Delta(\varepsilon, r)) - \alpha_0, \quad z_2(\Delta(\varepsilon, r)) - \beta_0 \quad (3.4)$$

are $O(e^{-C/\varepsilon})$ as $\varepsilon \rightarrow 0$, uniformly for $r \in \mathcal{R}_\delta$.

In Section 7 we will show that $\varepsilon_\delta \rightarrow 0$ as $\delta \rightarrow 0$.

We may assume without loss in generality that ε_δ is increasing for $0 < \delta < (\beta_0 - \alpha_0)/2$, so that $\Delta(\varepsilon, r)$ is well defined on the set

$$\mathcal{M} = \cup (0, \varepsilon_\delta) \times \mathcal{R}_\delta \text{ (union over } 0 < \delta < \tfrac{1}{2}(\beta_0 - \alpha_0)). \quad (3.5)$$

Given $(\varepsilon, r) \in \mathcal{M}$, we will refer to $\Delta(\varepsilon, r)$ as the *Maxwell solution* for (ε, r) .

In Section 8 we will show that the Maxwell solution corresponds to the global minimizer for our original problem, $\mathcal{P}_{\varepsilon r}$.

Remark. It is clear from the phase portraits that for $\alpha_0 < r < \beta_0$ all solutions of $\mathcal{S}_{\varepsilon r}$ lie on closed orbits. In view of the boundary conditions (2.4), such solutions begin and end with $\dot{z} = 0$. Let us agree to use the term *transitions* for times $t \neq \varepsilon^{-1}$ at which $\dot{z}(t) = 0$. With this definition a simple solution has a single transition. A trivial extension of the argument of Section 2 makes it clear that given any solution $z(t)$ with N transitions, there is a corresponding admissible Δ such that

$$I_0(\Delta) = 2N\varepsilon^{-1}, \quad I_1(\Delta) = 2Nr\varepsilon^{-1},$$

and conversely. We therefore arrive at an obvious analog of Theorem 3.1 which asserts, in particular, that for fixed N_0 and $r \in (\alpha_0, \beta_0)$, there are solutions with N transitions, $1 \leq N \leq N_0$, for all sufficiently small ε . We will not study these solutions in detail, because, as we shall see in Section 7, solutions with more than one transition cannot minimize energy.

A careful proof of Theorem 3.1 will be the subject of the next three sections. Here to motivate the analysis, which is quite complicated, we will sketch the proof.

By Theorem 2.1, the proof consists in solving the integral equations (2.21) near $\Delta = \Delta_0$. The main contribution to the integral $I_0(\Delta)$ comes from the singularities at $z = z_1(\Delta)$, $z_2(\Delta)$. If $T_1(\Delta)$ and $T_2(\Delta)$ denote the "contributions" from $z_1(\Delta)$ and $z_2(\Delta)$ to $I_0(\Delta)$, then

$$I_0(\Delta) \approx T_1(\Delta) + T_2(\Delta).$$

Since the integrand in $I_1(\Delta)$ weights these singularities by $z_1(\Delta)$ and $z_2(\Delta)$, while $z_1(\Delta) \approx \alpha_0$ and $z_2(\Delta) \approx \beta_0$,

$$I_1(\Delta) \approx \alpha_0 T_1(\Delta) + \beta_0 T_2(\Delta).$$

Thus to satisfy (2.21),

$$\begin{aligned} T_1(\Delta) &\approx \frac{2(\beta_0 - r)}{\varepsilon(\beta_0 - \alpha_0)}, \\ T_2(\Delta) &\approx \frac{2(r - \alpha_0)}{\varepsilon(\beta_0 - \alpha_0)}. \end{aligned} \tag{3.6}$$

Note that, since $T_i(\Delta) > 0$, (3.6) can hold only when r satisfies

$$\alpha_0 < r < \beta_0.$$

To solve (3.6) and hence (2.21), we need to know how $T_i(\Delta)$ depends on Δ . Here we give only the approximate results:

$$T_i(\Delta) \approx -B_i \ln h_i, \tag{3.7}$$

where

$$\begin{aligned} h_1 &= b - \Phi_\sigma(\alpha_\sigma), \\ h_2 &= b - \Phi_\sigma(\beta_\sigma), \end{aligned} \tag{3.8}$$

while

$$B_1 = [2W'''(\alpha_0)]^{-\frac{1}{2}}, \quad B_2 = [2W'''(\beta_0)]^{-\frac{1}{2}}. \quad (3.9)$$

We emphasize that (3.7) are *approximate*, as many terms have been neglected. If we substitute (3.7) into (3.6) and solve for h_i , we find that

$$h_i \approx \mathcal{K}_i e^{-c_i(r)/\varepsilon}$$

with $\mathcal{K}_i$ constants independent of ε and

$$\begin{aligned} c_1(r) &= \frac{2(\beta_0 - r)}{B_1(\beta_0 - \alpha_0)}, \\ c_2(r) &= \frac{2(r - \alpha_0)}{B_2(\beta_0 - \alpha_0)}. \end{aligned} \quad (3.10)$$

This crude calculation motivates introducing new parameters $k = (k_1, k_2)$ defined implicitly by

$$h_i = e^{\mu_i k_i} e^{-c_i(r)/\varepsilon}, \quad (3.11)$$

with μ_i constants chosen, for convenience, as

$$\mu_i = [B_i(\beta_0 - \alpha_0)]^{-1}. \quad (3.12)$$

We will show that with h_i given by (3.11), equation (2.21) has the form

$$k = F(k, \varepsilon, r), \quad (3.13)$$

where $F = (F_1, F_2)$ is a C^1 function whose derivative with respect to k vanishes at $\varepsilon = 0$. We will use the implicit function theorem to solve (3.13) and hence (2.21).

Most of our analysis concerns behavior near the Maxwell point Δ_0 . In Section 4 we will study the transformation $h = \pi(\Delta)$ defined by (3.8), and we will show that $z_1(\Delta) - \alpha_\sigma$ and $\beta_\sigma - z_2(\Delta)$, respectively, behave like $h_1^{\frac{1}{2}}$ and $h_2^{\frac{1}{2}}$ near Δ_0 . (Note that $h = 0$ at $\Delta = \Delta_0$.)

Section 5 is concerned, for the most part, with the behavior of the integrals $I_n(\Delta)$ near $\Delta = \Delta_0$. Here we will also establish certain estimates away from Δ_0 ; these are needed in Sections 7 and 8, where we show that the Maxwell solution actually minimizes energy.

In Section 6 we prove Theorem 3.1, using as a basis the scaling (3.11). There we will reduce the integral equations (3.21) to equations of the form (3.13). This will necessitate a careful study of $I_n(\Delta)$ —for Δ a function of (k, ε, r) —near $\varepsilon = 0$.

4. Properties of $z_i(\Delta)$. The mapping $h = \pi(\Delta)$

We begin by listing several important properties of the critical points of the Gibbs function.

Proposition 4.1. *The critical points α_σ , ζ_σ , and β_σ , as functions of σ , are continuous on $[\underline{\sigma}, \bar{\sigma}]$ and C^4 on $(\underline{\sigma}, \bar{\sigma})$ with α_σ and β_σ strictly increasing, ζ_σ strictly decreasing. Further,*

$$\alpha_{\bar{\sigma}} = \zeta_{\bar{\sigma}} = \bar{\alpha}, \quad \beta_{\underline{\sigma}} = \zeta_{\underline{\sigma}} = \underline{\beta}, \quad (4.1)$$

and for $\underline{\sigma} < \sigma < \bar{\sigma}$,

$$W''(\alpha_\sigma), W''(\beta_\sigma) > 0, \quad W''(\zeta_\sigma) < 0. \quad (4.2)$$

Proof. Let ξ_σ denote α_σ , ζ_σ , or β_σ . The smoothness of $\sigma \mapsto \xi_\sigma$ as well as (4.1) and (4.2) follow from the properties of W' in conjunction with the implicit relation (2.12) defining ξ_σ . Further, (2.12) implies that

$$W''(\xi_\sigma) \frac{d\xi_\sigma}{d\sigma} = 1, \quad (4.3)$$

which, with (4.2), leads to the asserted monotonicity conditions. $\square$

We now use these results to study the functions $z_i(\Delta)$ over their domain

$$\Sigma = \text{the set of admissible pairs}. \quad (4.4)$$

For convenience, we write $\partial\Sigma$ as the disjoint union

$$\partial\Sigma = \bigcup_{i=1}^3 \partial_i\Sigma,$$

where

$$\begin{aligned} \partial_1\Sigma &= \{(\sigma, b) : b = \Phi_\sigma(\alpha_\sigma), \sigma_0 \leq \sigma < \bar{\sigma}\}, \\ \partial_2\Sigma &= \{(\sigma, b) : b = \Phi_\sigma(\beta_\sigma), \underline{\sigma} < \sigma \leq \sigma_0\}, \\ \partial_3\Sigma &= \{(\sigma, b) : b = \Phi_\sigma(\zeta_\sigma), \underline{\sigma} \leq \sigma \leq \bar{\sigma}\}. \end{aligned} \quad (4.5)$$

Note that $\partial_1\Sigma$ and $\partial_2\Sigma$ intersect at Δ_0 (cf. (3.1)). $\partial\Sigma$ also has corners (actually cusps) at the points

$$\Delta = (\underline{\sigma}, \underline{b}), \quad \bar{\Delta} = (\bar{\sigma}, \bar{b}), \quad (4.6)$$

where

$$\underline{b} = \Phi_{\underline{\sigma}}(\underline{\beta}), \quad \bar{b} = \Phi_{\bar{\sigma}}(\bar{\alpha}).$$

Proposition 4.2. *$z_i(\Delta)$ is continuous on $\bar{\Sigma}$ and C^4 on Σ . Moreover, for $\Delta \in \Sigma$,*

$$\alpha_0 < z_1(\Delta) < z_2(\Delta) < \beta_0, \quad (4.7)$$

while

$$z_1(\Delta) = \alpha_\sigma \text{ for } \Delta \in \partial_1\Sigma, \quad z_2(\Delta) = \beta_\sigma \text{ for } \Delta \in \partial_2\Sigma, \quad (4.8)$$

$$z_1(\Delta) = z_2(\Delta) = \zeta_\sigma \text{ for } \Delta \in \partial_3\Sigma,$$

and

$$\begin{aligned} \Phi'_\sigma(z_1(\Delta)) &\neq 0 \text{ for } \Delta \in \partial_2\Sigma \setminus \{\Delta_0\}, \\ \Phi'_\sigma(z_2(\Delta)) &\neq 0 \text{ for } \Delta \in \partial_1\Sigma \setminus \{\Delta_0\}. \end{aligned} \quad (4.9)$$

Proof. The result (4.8) follows from (4.6). To verify (4.9) choose $\Delta \in \partial_1 \Sigma \setminus \{\Delta_0\}$. If $z_2(\Delta) = \zeta_\sigma$, then $z_1(\Delta) = \zeta_\sigma$, which, with (4.8)₁, implies $\sigma = \bar{\sigma}$, an impossibility when $\Delta \in \partial_1 \Sigma$. On the other hand, for $z_2(\Delta) = \beta_\sigma$, (4.8)₁ yields $\Phi_\sigma(\alpha_\sigma) = \Phi_\sigma(\beta_\sigma)$, a possibility only for $\sigma = \sigma_0$ (cf. (iii) of Proposition 2.1). Thus (4.9)₂ holds. A similar argument yields (4.9)₁.

We now turn to (4.7) and the assertions of regularity, which we shall establish only for $z_1(\Delta)$.

The function g_σ in (2.19) is well defined at all $\sigma \in [\underline{\sigma}, \bar{\sigma}]$ except $\sigma = \bar{\sigma}$. Thus (2.15), (2.19), the smoothness of the Gibbs function, Proposition 3.1, and the fact that $\Phi'_\sigma > 0$ on $(\alpha_\sigma, \zeta_\sigma)$ imply that $z_1(\Delta)$ is C^4 on Σ and continuous at all points of $\bar{\Sigma}$ except possibly the boundary point $\bar{\Delta}$ corresponding to $\bar{\sigma}$. That $z_1(\Delta)$ is also continuous at $\bar{\Delta}$ follows from (4.1)₁ and the inequality $\alpha_\sigma < z_1(\Delta) < \beta_\sigma$, $\Delta \in \Sigma$.

Our next step will be to show that $\alpha_0 < z_1(\Delta)$. Assume that $\sigma < \sigma \leq \sigma_0$. Since $\Phi_\sigma(z)$ is strictly increasing in z on $[\alpha_\sigma, \zeta_\sigma]$, since both α_0 and $z_1(\Delta)$ belong to this interval, and since (cf. Figure 2) $\Phi_\sigma(\beta_\sigma) < \Phi_\sigma(z_1(\Delta))$, it suffices to show that

$$\Phi_\sigma(\alpha_0) \leq \Phi_\sigma(\beta_\sigma). \quad (4.10)$$

Since

$$\frac{\partial}{\partial \sigma} \Phi_\sigma(z) = -z, \quad (4.11)$$

we may use (2.13) to conclude that

$$\frac{d}{d\sigma} [\Phi_\sigma(\beta_\sigma) - \Phi_\sigma(\alpha_0)] = \alpha_0 - \beta_\sigma < 0,$$

and this inequality, with (2.11), leads to (4.10) for $\sigma \leq \sigma_0$. Further, $z_1(\Delta) < \beta_\sigma$ and the monotonicity of β_σ insures that $z_1(\Delta) \leq \beta_0$ for $\sigma \leq \sigma_0$. Hence $\alpha_0 < z_1(\Delta) < \beta_0$ for $\Delta \in \Sigma$ with $\sigma \leq \sigma_0$. An analogous argument yields this inequality for $\sigma > \sigma_0$. $\square$

Our analysis is greatly simplified if, in place of $\Delta = (\sigma, b)$, we use the parameters

$$\begin{aligned} h_1 &= b - \Phi_\sigma(\alpha_\sigma), \\ h_2 &= b - \Phi_\sigma(\beta_\sigma). \end{aligned} \quad (4.12)$$

Writing

$$h = (h_1, h_2), \quad (4.13)$$

we denote the transformation (4.12) by

$$h = \pi(\Delta) \quad (h_i = \pi_i(\Delta)).$$

In view of (4.8),

$$h_2 - h_1 = \Phi_\sigma(\alpha_\sigma) - \Phi_\sigma(\beta_\sigma) \equiv \eta(\sigma) \quad (4.14)$$

and since (2.13) yields

$$\frac{d\eta(\sigma)}{d\sigma} = \beta_\sigma - \alpha_\sigma > 0,$$

$\underline{\sigma} \leq \sigma \leq \bar{\sigma}$, it is clear that π is an isomorphism of $\bar{\Sigma}$ onto the closure of the set

$$\mathcal{H} = \pi(\Sigma).$$

Further, a simple calculation using (2.13) shows that the Jacobian determinant of the mapping π is

$$\det \begin{bmatrix} \alpha_\sigma & 1 \\ \beta_\sigma & 1 \end{bmatrix} = \alpha_\sigma - \beta_\sigma < 0,$$

$\underline{\sigma} \leq \sigma \leq \bar{\sigma}$; hence π is a C^4 diffeomorphism of $\bar{\Sigma}$ onto $\bar{\mathcal{H}}$.

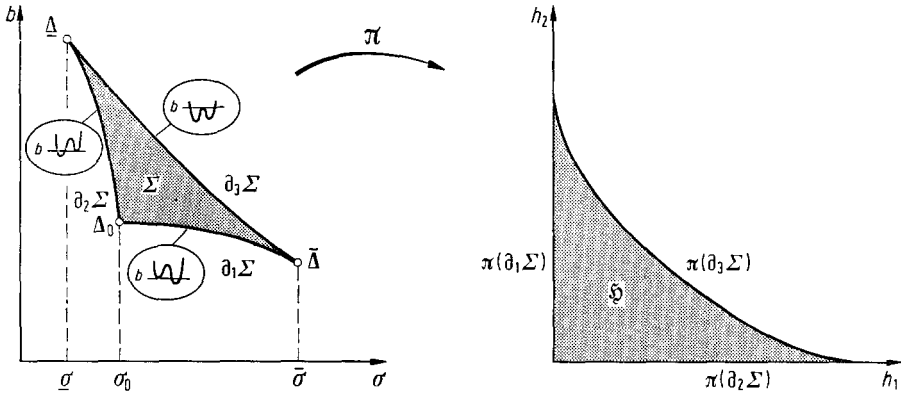


Fig. 3. The sets Σ and $\mathcal{H} = \pi(\Sigma)$

Note that, by (3.1), (4.1), (4.6), and (4.12),

$$\pi(\Delta_0) = 0, \quad \pi_1(\bar{\Delta}) = 0, \quad \pi_2(\underline{\Delta}) = 0. \quad (4.15)$$

Much of our analysis concerns situations with Δ close to Δ_0 , or equivalently, situations with h close to $h = 0$. With this in mind, we say that a function $\psi(h)$ is **nearly regular at** $h = 0$ if $\psi(h)$ is C^1 for sufficiently small $h \in \mathcal{H}$ and if, for some $s < 1$,

$$\begin{aligned} \psi(h) &= \psi(0) + O(|h|^{1-s}), \\ \nabla \psi(h) &= (O(h_1^{-s}), O(h_2^{-s})) \end{aligned} \quad (4.16)$$

as $h \rightarrow 0$, $h \in \mathcal{H}$.

Given a function $\varphi(\Delta)$, we write

$$\hat{\varphi}(h) = \varphi(\pi^{-1}(h)). \quad (4.17)$$

We will also use the following notation: for $\Omega \subset \mathbb{R}^2$,

$$C^+(\Omega) = \{f \in C(\Omega) : f > 0 \text{ on } \Omega\};$$

for Ω a neighborhood of Δ_0 in $\bar{\Sigma}$,

$$NR(\Omega) = \{\varphi \in C(\Omega) : \hat{\varphi}(h) \text{ is nearly regular at } h = 0\},$$

$$NR^+(\Omega) = \{\varphi \in NR(\Omega) : \varphi > 0 \text{ on } \Omega\}.$$

Proposition 4.3. *The functions $z_i(\Delta)$ belong to $NR(\bar{\Sigma})$. In fact, for $h = \pi(\Delta)$,*

$$\begin{aligned} z_1(\Delta) &= \alpha_\sigma + h_1^{\frac{1}{2}} \Theta_1(\Delta), \\ z_2(\Delta) &= \beta_\sigma - h_2^{\frac{1}{2}} \Theta_2(\Delta) \end{aligned} \quad (4.18)$$

on Σ with

$$\Theta_i \in NR^+(\bar{\Sigma} \setminus \{\underline{\Delta}, \bar{\Delta}\}). \quad (4.19)$$

Lemma 4.1. *Let $G(x, p)$ be a C^1 function on a neighborhood of $0 \in \mathbb{R}^2$. Then for all sufficiently small (p, q) the equation*

$$x^2 + x^3 G(x, p) = q^2 \quad (4.20)$$

has a solution

$$x(p, q) = qy(p, q), \quad (4.21)$$

where $y(p, q)$ is C^1 and satisfies

$$y(0, 0) = 1. \quad (4.22)$$

Proof. If we set $x = qy$, (4.20) becomes

$$F(y, p, q) = y^2 + qy^3 G(qy, p) - 1 = 0 \quad (4.23)$$

with

$$F(1, 0, 0) = 0, \quad \frac{\partial F}{\partial y}(1, 0, 0) = 2.$$

We may therefore conclude from the implicit function theorem that for (p, q) sufficiently small (4.23) has a C^1 solution $y = y(p, q)$ with $y(0, 0) = 1$. $\square$

Proof of Proposition 4.3. We shall establish this proof only for $z_1(\Delta)$. The proof for $z_2(\Delta)$ is identical.

Since $h_1 > 0$ and $\alpha_\sigma < z_1(\Delta)$ in $\bar{\Sigma} \setminus (\partial_1 \Sigma \cup \{\bar{\Delta}\})$ (cf. (4.9)), (4.18) defines $\Theta_1(\Delta)$ as a $C^+(\partial_1 \Sigma \cup \{\bar{\Delta}\})$ function. To complete the proof we must show that $\hat{\Theta}_1$ satisfies (4.16), and that given an arbitrary point $\Delta_1 = (\sigma_1, b_1) \in \partial_1 \Sigma$,

$$\Theta_1(\Delta) \text{ is continuous at } \Delta_1 \text{ and } \Theta_1(\Delta_1) > 0. \quad (4.24)$$

Clearly,

$$t = t(\Delta) = z_1(\Delta) - \alpha_\sigma$$

is the unique solution, in $[0, \lambda_\sigma - \alpha_\sigma]$, of the equation

$$f_\Delta(\alpha_\sigma + t) = 0. \quad (4.25)$$

By (2.7) and (2.17),

$$\begin{aligned} f'_\Delta(z) &= \Phi'_\Delta(z), \\ f_\Delta^{(n)}(z) &= \Phi_\Delta^{(n)}(z) = W^{(n)}(z), \quad n = 2, 3, 4; \end{aligned} \quad (4.26)$$

we may therefore conclude from Taylor's formula with remainder, (2.10), and (2.17) that

$$f_{\Delta}(\alpha_{\sigma} + t) = \frac{1}{2}W''(\alpha_{\sigma})t^2 + g(t, \sigma)t^3 + \Phi_{\sigma}(\alpha_{\sigma}) - b$$

with $g(t, \sigma)$ a C^1 function on a neighborhood of the point $(0, \sigma_1)$. Thus if we let

$$x = [W''(\alpha_{\sigma})/2]^{\frac{1}{2}}t,$$

$$p = \sigma - \sigma_1,$$

$$q = [b - \Phi_{\sigma}(\alpha_{\sigma})]^{\frac{1}{2}} = h_1^{\frac{1}{2}}$$

(cf. (4.2)), then (4.25) reduces to (4.20). Let $x(p, q)$ be the solution (4.21), (4.22). Then tracing back the above transformations, we conclude that for all $\Delta \in \Sigma$ which lie sufficiently close to Δ_1 ,

$$\Theta(\Delta) = [2/W''(\alpha_{\sigma})]^{\frac{1}{2}}y(\sigma - \sigma_1, h_1^{\frac{1}{2}}).$$

Because of (4.22) and the smoothness of y , this in turn leads to (4.24) and, when $\sigma_1 = \sigma_0$, to (4.16) with $\varphi = \hat{\Theta}_1$ and $s = \frac{1}{2}$. $\square$

Remark. Actually, $z_1(\Delta)$ and $z_2(\Delta)$ obey the estimates

$$\nabla \hat{z}_1(h) = (O(h_1^{-\frac{1}{2}}), O(h_1^{\frac{1}{2}})),$$

$$\nabla \hat{z}_2(h) = (O(h_2^{\frac{1}{2}}), O(h_2^{-\frac{1}{2}}))$$

as $h \rightarrow 0$, $h \in \mathcal{H}$.

We close this section with a lemma which will be useful in our analysis of the integrals $I_n(\Delta)$. Its proof is an obvious consequence of Proposition 4.3 and the chain-rule.

Lemma 4.2. *Let*

$$\varphi(\Delta) = F(\Delta, z_1(\Delta), z_2(\Delta))$$

with F a C^1 function on

$$\{(\Delta, y_1, y_2) : \Delta \in \bar{\Sigma} \cap \mathcal{B}_{\delta}, z_1(\Delta) \leq y_1 \leq z_1(\Delta) + \delta, z_2(\Delta) - \delta \leq y_2 \leq z_2(\Delta)\},$$

where $\delta > 0$ and $\mathcal{B}_{\delta}$ is the disc of radius δ centered at Δ_0 . Then $\varphi \in NR(\bar{\Sigma} \cap \mathcal{B}_{\delta})$.

5. Properties of $I_n(\Delta)$

Let

$$\mathcal{W}_i = \{\Delta \in \bar{\Sigma} : W''(z_i(\Delta)) > 0\},$$

and let Ω_0 , Ω_1 , Ω_2 be (arbitrary) closed subregions of $\bar{\Sigma}$ with (Figure 4)

$$\Omega_0 \subset \mathcal{W}_1 \cap \mathcal{W}_2,$$

$$\Omega_1 \subset (\mathcal{W}_1 \setminus \partial_2 \Sigma), \tag{5.1}$$

$$\Omega_2 \subset (\mathcal{W}_2 \setminus \partial_1 \Sigma),$$

so that

$$W'''(z_i(\Delta)) \geq C > 0 \text{ on } \Omega_i \cup \Omega_0. \quad (5.2)$$

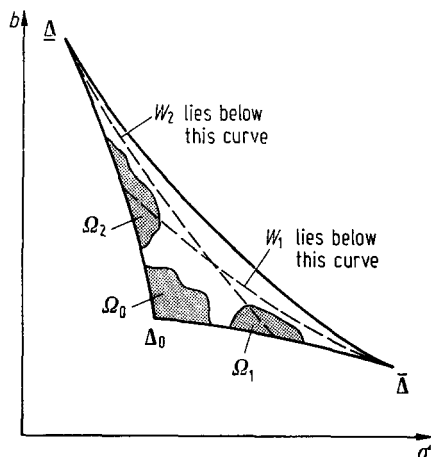


Fig. 4. The sets Ω_0 , Ω_1 , Ω_2 . Σ is the master set

The integral (2.20) defining $I_n(\Delta)$ has an integrand which is singular at $z = z_i(\Delta)$. We therefore choose $\varkappa > 0$ and write

$$I_n = J_n + K_n + L_n, \quad (5.3)$$

where

$$\begin{aligned} J_n(\Delta) &= \int_{z_1(\Delta)}^{z_1(\Delta)+\varkappa} z^n f(z)^{-\frac{1}{2}} dz = \int_0^\varkappa (z_1(\Delta) + t)^n f_\Delta(z(\Delta) + t)^{-\frac{1}{2}} dt, \\ K_n(\Delta) &= \int_{z_2(\Delta)-\varkappa}^{z_2(\Delta)} z^n f_\Delta(z)^{-\frac{1}{2}} dz = \int_0^\varkappa (z_2(\Delta) - t)^n f_\Delta(z_2(\Delta) - t)^{-\frac{1}{2}} dt, \\ L_n(\Delta) &= \int_{z_1(\Delta)+\varkappa}^{z_2(\Delta)-\varkappa} z^n f_\Delta(z)^{-\frac{1}{2}} dz. \end{aligned} \quad (5.4)$$

Note that, by (4.8), (4.9), and Proposition 2.2, $z_1(\Delta) < \zeta_\sigma < z_2(\Delta)$ on the closed set $\Omega_0 \cup \Omega_1 \cup \Omega_2$; thus on that set $z_1(\Delta) + \varkappa < z_2(\Delta)$ for all sufficiently small $\varkappa$.

Since $f_\Delta(z_i(\Delta)) = 0$, while $f'_\Delta(z) = \Phi'_\sigma(z) = 0$ at $z = \alpha_\sigma, \beta_\sigma$, it follows that

$$\begin{aligned} f_\Delta(z_1(\Delta) + t)^{\frac{1}{2}} &\sim Ct^{\frac{1}{2}} \text{ for } z_1(\Delta) > \alpha_\sigma, \\ f_\Delta(z_1(\Delta) + t)^{\frac{1}{2}} &\sim Ct \text{ for } z_1(\Delta) = \alpha_\sigma, \end{aligned}$$

and $J_n(\Delta)$ is continuous away from—but singular at—points Δ with $z_1(\Delta) = \alpha_\sigma$. A similar assertion applies to $K_n(\Delta)$. Thus, by (4.8),

$$J_n \in C(\Omega_2), \quad K_n \in C(\Omega_1), \quad L_n \in NR(\Omega_0 \cup \Omega_1 \cup \Omega_2) \quad (5.5)$$

(cf. Lemma 4.2). As we shall see, $J_n(\Delta)$ behaves like $\ln h_1$ near $\partial_1 \Sigma$, while $K_n(\Delta)$ behaves like $\ln h_2$ near $\partial_2 \Sigma$. The precise nature of these singularities is given in the next proposition; there

$$Q_i(\Delta) = [2W'''(z_i(\Delta))]^{-\frac{1}{2}}, \quad (5.6)$$

so that, by (5.2) and Lemma 4.2,

$$Q_i \in NR^+(\Omega_i \cup \Omega_0). \quad (5.7)$$

Proposition 5.1. For $n = 0, 1$,

$$\begin{aligned} I_n + Q_1 z_1^n \ln \pi_1 &\in C(\Omega_1), \quad I_n + Q_2 z_2^n \ln \pi_2 \in C(\Omega_2), \\ I_n + Q_1 z_1^n \ln \pi_1 + Q_2 z_2^n \ln \pi_2 &\in NR(\Omega_0). \end{aligned} \quad (5.8)$$

Proof. In view of (5.3) and (5.5), it suffices to show that

$$J_n + Q_1 z_1^n \ln \pi_1 \in NR(\Omega_1 \cup \Omega_0), \quad K_n + Q_2 z_2^n \ln \pi_2 \in NR(\Omega_2 \cup \Omega_0). \quad (5.9)$$

We shall prove only (5.9)₁; the proof of (5.9)₂ is analogous. Thus, since (5.9)₁ involves only $z_1(\Delta)$ with Δ confined to $\Omega_1 \cup \Omega_0$, we shall use the simplified notation

$$z(\Delta) = z_1(\Delta),$$

$$Q(\Delta) = Q_1(\Delta),$$

$$\Omega = \Omega_1 \cup \Omega_0$$

for the remainder of the proof.

If we expand $f_\Delta(y+t)$ about $t=0$ using Taylor's formula with remainder in conjunction with (4.26), we find that

$$f_\Delta(y+t) = f_\Delta(y) + \Phi'_\sigma(y)t + \frac{1}{2}W'''(y)t^2 + \frac{1}{6}W''''(y)t^3 + \mathcal{R}(y,t)t^4$$

with $\mathcal{R}(y,t)$ a C^1 function for $y, y+t > 0$. Thus, since

$$\Phi'_\sigma(y) = W'(y) - \sigma = W'(y) - W'(\alpha_\sigma),$$

if we take $y = z(\Delta)$ we find that

$$f_\Delta(z(\Delta) + t) = [2Q(\Delta)]^{-2} R(t, \Delta), \quad (5.10)$$

where

$$\begin{aligned} R(t, \Delta) &= t^2 + 2\lambda(\Delta)t + c(\Delta)t^3 + g(t, \Delta)t^4, \\ \lambda(\Delta) &= [W'(z(\Delta)) - W'(\alpha_\sigma)]/W'''(z(\Delta)), \end{aligned} \quad (5.11)$$

$$c(\Delta) = W''''(z(\Delta))/3W'''(z(\Delta)), \quad g(t, \Delta) = 2\mathcal{R}(z(\Delta), t)/W'''(z(\Delta)).$$

Note that

$$\lambda, c \in C(\Omega), \quad g \in C([0, \infty) \times \Omega). \quad (5.12)$$

In addition, by (5.2) (cf. Figure 2),

$$\lambda > 0 \quad \text{on } \Omega \setminus \partial_1 \Sigma. \quad (5.13)$$

We now use (5.10) to rewrite the integral (5.4)₁ as

$$J_n(\Delta) = 2Q(\Delta) \int_0^\infty [z_1(\Delta) + t]^n R(t, \Delta)^{-\frac{1}{2}} dt. \quad (5.14)$$

The term $2\lambda(\Delta)t$ in the expression for $R(t, \Delta)$ leads to terms involving $\ln \lambda$ and $\lambda \ln \lambda$ when estimating the integral J_n . For this reason we now establish

Lemma 5.1.

$$\ln \lambda - \frac{1}{2} \ln \pi_1 \in NR(\Omega), \quad \lambda \ln \lambda \in NR(\Omega). \quad (5.15)$$

Proof. The function defined by

$$\begin{aligned} f(y, \Delta) &= \frac{W'(y) - W'(\alpha_\sigma)}{(y - \alpha_\sigma) W''(y)}, \quad y \neq \alpha_\sigma \\ &= 1, \quad y = \alpha_\sigma \end{aligned}$$

is C^1 for all $\Delta \in \Omega$ and y with $W''(y) > 0$. Thus if we define

$$\Delta_0(\Delta) = f(z(\Delta), \Delta),$$

appeal to the definition of $\lambda(\Delta)$ in (5.11), and use Lemma 4.2, (5.12), and (5.13), we find that

$$\lambda(\Delta) = [z(\Delta) - \alpha_\sigma] \Delta_0(\Delta)$$

on Ω with $\Delta_0 \in NR^+(\Omega)$. Therefore, by Proposition 4.3,

$$\lambda(\Delta) = \pi_1(\Delta)^{\frac{1}{2}} \Delta(\Delta)$$

with $\Delta \in NR^+(\Omega)$, so that $\ln \Delta \in NR(\Omega)$. Thus (5.15)₁ is valid.

To prove that $\lambda \ln \lambda \in NR(\Omega)$, it clearly suffices to show that $h_1^{\frac{1}{2}} \ln h_1$ is nearly regular at $h = 0$. But this is obvious. $\square$

The next lemma gives the precise behavior of the integral (5.14).

Lemma 5.2. *There is a $\varkappa > 0$ such that*

$$\int_0^\varkappa R(t, \Delta)^{-\frac{1}{2}} dt + \ln \lambda(\Delta) \quad \text{and} \quad \int_0^\varkappa t R(t, \Delta)^{-\frac{1}{2}} dt \quad (5.16)$$

belong to $NR(\Omega)$.

Proof. We begin by writing

$$R(t, \Delta) = [t^2 + 2\lambda(\Delta)t][1 + K(t, \Delta)],$$

$$K(t, \Delta) = [c(\Delta)t^3 + g(t, \Delta)t^4]/[t^2 + 2\lambda(\Delta)t].$$

In view of (5.12) and (5.13), we can choose $\varkappa$ (independent of Δ) such that $|K(t, \Delta)| \leq \frac{1}{2}$ for $t \in [0, \varkappa]$, $\Delta \in \Omega$. By the binomial theorem,

$$[1 + K(t, \Delta)]^{-\frac{1}{2}} = 1 - \frac{1}{2}K(t, \Delta) + \sum_{m=2}^{\infty} a_m K(t, \Delta)^m,$$

with a_m appropriate binomial coefficients. Thus

$$R(t, \Delta)^{-\frac{1}{2}} = [t^2 + 2\lambda(\Delta)t]^{-\frac{1}{2}} - \frac{1}{2}c(\Delta)t^3[t^2 + 2\lambda(\Delta)t]^{-\frac{3}{2}} \\ - \frac{1}{2}g(t, \Delta)t^4[t^2 + 2\lambda(\Delta)t]^{-\frac{5}{2}} + H(t, \Delta) \quad (5.17)$$

with

$$H(t, \Delta) = \sum_{m=2}^{\infty} a_m H_m(t, \Delta), \\ H_m(t, \Delta) = [c(\Delta)t^3 + g(t, \Delta)t^4]^m/[t^2 + 2\lambda(\Delta)t]^{m+\frac{1}{2}}. \quad (5.18)$$

Routine integrations show that

$$\int (t^2 + 2\lambda t)^{-\frac{1}{2}} dt = \ln [t + \lambda + (t^2 + 2\lambda t)^{\frac{1}{2}}], \\ \int t^3(t^2 + 2\lambda t)^{-\frac{3}{2}} dt = 3(t^2 + 2\lambda t)^{\frac{1}{2}} - 3\lambda \ln [t + \lambda + (t^2 + 2\lambda t)^{\frac{1}{2}}] \\ - 2t^2(t^2 + 2\lambda t)^{-\frac{1}{2}}. \quad (5.19)$$

Thus and by Lemma 5.1, the contributions of the first two terms in (5.17) to the integral in (5.16)₁ is

$$-\ln \lambda(\Delta) + \text{a function in } NR(\Omega).$$

Consider next the integral of $H(t, \Delta)$. In view of (5.11),

$$g(t, \Delta) = \tilde{g}(t, z(\Delta)), \quad c(\Delta) = \tilde{c}(z(\Delta)), \quad \lambda(\Delta) = \tilde{\lambda}(\Delta, z(\Delta))$$

with $\tilde{g}(t, y)$, $\tilde{c}(y)$, and $\tilde{\lambda}(\Delta, y)$ C^1 for $\Delta \in \Omega$, $t \in [0, \kappa]$, and $y \in z(\Omega)$ (as $W''(y) > 0$ for $y \in z(\Omega)$). Thus by (5.18),

$$H(t, \Delta) = \tilde{H}(t, \Delta, z(\Delta)) = \sum_{m=2}^{\infty} a_m \tilde{H}_m(t, \Delta, z(\Delta))$$

with $\tilde{H}_m(t, \Delta, z(\Delta))$ defined in the obvious manner; i.e., by (5.18)₂ with $\lambda(\Delta)$ replaced by $\tilde{\lambda}(\Delta, z(\Delta))$, etc. For $m \geq 2$, $\tilde{H}_m(t, \Delta, y)$ is $O(t^{m-1})$ and hence continuous for $t \in [0, \kappa]$, $\Delta \in \Omega$, $y \in z(\Omega)$, and this, in turn, leads to the same degree of continuity for $\tilde{H}(t, \Delta, y)$. Further, a simple calculation shows that, for κ sufficiently small, there is a constant $C > 0$, independent of t, Δ, y , and m , such that

$$\frac{\partial}{\partial y} \tilde{H}_m(t, \Delta, y) \quad (5.20)$$

is bounded in absolute value by mCt^{m-2} for $t \in [0, \kappa]$, $\Delta \in \Omega$, $y \in z(\Omega)$, $m \geq 2$. Moreover, for $m \geq 3$, (5.20) is continuous for all such (t, Δ, y) , while for $m = 2$, (5.20) is continuous in (Δ, y) for each t . Thus

$$\int_0^{\kappa} \tilde{H}(t, \Delta, y) dt$$

belongs to $NR(\Omega)$.

A far easier analysis tells us that the contribution of the third term in (5.17) to the integral in (5.16)₁ is a function in $NR(\Omega)$. Thus (5.16)₁ is valid.

We omit the proof of (5.16)₂, which is similar. $\square$

We are now in a position to complete the proof of Proposition 5.1. Indeed, (5.14), (5.15), and (5.16) combine to give the desired estimate (5.9)₁. $\square$

Remark. An obvious corollary to Proposition 5.1 is that

$$I_n(\Delta) \rightarrow \infty \text{ as } \Delta \rightarrow \partial_1 \Sigma \cup \partial_2 \Sigma \quad (\Delta \in \Sigma). \quad (5.21)$$

Remark. Let Δ be admissible, and let $z(t)$ be the corresponding simple solution of $\mathcal{S}_{er}$. Then the integral (5.14; $n = 0$)—considered as a function $J_0(\Delta, \kappa)$ —gives the “time” taken by $z(t)$ to go from $z_1(\Delta)$ to $z_1(\Delta) + \kappa$. An analysis that closely follows the proof of Lemma 5.2 (*cf.*, in particular, (5.19)) establishes the existence of a $\kappa_0 > 0$ such that

$$J_0(\Delta, \kappa) = 2Q_1(\Delta) \{ \ln [\kappa + \lambda + (\kappa^2 + 2\lambda\kappa)^{\frac{1}{2}}] - \ln \lambda \} + \mathcal{J}(\Delta, \kappa) \quad (5.22)$$

with $\mathcal{J}$ continuous on $\Omega \times [0, \kappa_0]$ and $\lambda = \lambda(\Delta)$ given by (5.11). A similar interpretation and result applies to the integral $K_0(\Delta, \kappa)$ defined by (5.4)₂.

6. Proof of Theorem 3.1

In view of Theorem 2.1, finding a simple solution of $\mathcal{S}_{er}$ is equivalent to finding an admissible pair Δ that satisfies

$$\hat{\mathcal{S}}_{er} \begin{cases} I_0(\Delta) = 2e^{-1} \\ I_1(\Delta) = 2re^{-1}. \end{cases} \quad (2.21)$$

Our proof of Theorem 3.1 will consist in solving $\hat{\mathcal{S}}_{er}$ in a neighborhood of Δ_0 . With this in mind, and using as motivation the discussion given in Section 3, we introduce the scaling

$$h_i = e^{\mu_i k_i} e^{-c_i(r)/\varepsilon}, \quad (6.1)$$

$i = 1, 2$, where μ_i and $c_i(r)$ are defined by (3.10) and (3.12) with

$$c_i(r) > 0$$

for $r \in (\alpha_0, \beta_0)$. We will proceed as follows:

(i) We will show that sufficiently nice functions $\varphi(\Delta)$, when expressed as functions $\varphi^*(k, \varepsilon, r)$ of (k, ε, r) , can be extended smoothly to $\varepsilon \leq 0$.

(ii) Using this fact, we will reduce the integral equations (2.21) to an equation of the form $k = F(k, \varepsilon, r)$, which we will solve near $\varepsilon = 0$ using the implicit function theorem.

(iii) We will show that no solutions are lost in the scaling (6.1).

The transformation (6.1) is defined for $k = (k_1, k_2) \in \mathbb{R}^2$, $\varepsilon > 0$, and $r \in (\alpha_0, \beta_0)$. When h defined by (6.1) belongs to $\mathcal{H}$, we will call (k, ε, r) **compatible**;

in this instance we can compute a corresponding admissible pair through $\Delta = \pi^{-1}(h)$; we denote the resulting mapping by

$$\Delta = D(k, \varepsilon, r).$$

Conversely, given any $\Delta \in \Sigma$ and any $\varepsilon > 0$, we can define $h = \pi(\Delta)$ and solve (6.1) for k ; we denote this mapping by

$$k = K(\Delta, \varepsilon, r).$$

(Trivially, $\Delta \mapsto K(\Delta, \varepsilon, r)$ is the inverse of $k \mapsto D(k, \varepsilon, r)$.)

Given any function $\varphi(\Delta)$, we write

$$\varphi^*(k, \varepsilon, r) = \varphi(D(k, \varepsilon, r)).$$

The next lemma shows that, in certain circumstances, φ^* may be extended smoothly to a neighborhood of $\varepsilon = 0$; i.e., a neighborhood in $\mathbb{R}^4$ of the set

$$\mathcal{E}_0 = \{(k, \varepsilon, r) : k \in \mathbb{R}^2, \varepsilon = 0, r \in (\alpha_0, \beta_0)\}. \quad (6.2)$$

Let $\mathcal{K}$ be a neighborhood of $\varepsilon = 0$. Then we write

$$\mathcal{K}^+ = \{(k, \varepsilon, r) \in \mathcal{K} : \varepsilon > 0\}$$

and call $\mathcal{K}$ **compatible** if every $(k, \varepsilon, r) \in \mathcal{K}^+$ is compatible. Finally, given any C^1 function $\Phi(k, \varepsilon, r)$, we write

$$\nabla \Phi = \left(\frac{\partial \Phi}{\partial k_1}, \frac{\partial \Phi}{\partial k_2}, \frac{\partial \Phi}{\partial \varepsilon}, \frac{\partial \Phi}{\partial r} \right).$$

Let Ω_0 be a closed subregion of $\bar{\Sigma}$ consistent with (5.1).

Lemma 6.1. *Let*

$$\varphi = \varphi_0 + \varphi_1 \ln \pi_1 + \varphi_2 \ln \pi_2$$

with

$$\varphi_0, \varphi_1, \varphi_2 \in NR(\Omega_0), \quad \varphi_1(\Delta_0) = \varphi_2(\Delta_0) = 0.$$

Then there is a compatible neighborhood $\mathcal{K}$ of $\mathcal{E}_0$ such that

- (i) φ^* has a C^1 extension to $\mathcal{K}$;
- (ii) $\nabla \varphi^* = 0$ on $\mathcal{E}_0$.

Proof. By hypothesis and the definition of a nearly regular function, $\varphi^*(k, \varepsilon, r)$ is C^1 on a set of the form

$$\mathcal{K}^* = \{(k, \varepsilon, r) : h \in \mathcal{H}, \text{ defined by (6.1), is sufficiently small}\}.$$

We take

$$\mathcal{K} = \mathcal{K}^* \cup \{(k, \varepsilon, r) : k \in \mathbb{R}^2, \varepsilon \leq 0, r \in (\alpha_0, \beta_0)\},$$

which is clearly a compatible neighborhood of $\mathcal{E}_0$.

To prove the lemma it suffices to show that, for $k \in \mathbb{R}^2$ and $r \in (\alpha_0, \beta_0)$,

$$\nabla \varphi^*(k, \varepsilon, r) \rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0. \quad (6.3)$$

Indeed, (6.3) implies that φ^* is C^1 up to $\varepsilon = 0$ with

$$\nabla \varphi^* = 0 \text{ on } \mathcal{E}_0,$$

and this implies that $\varphi^*(k, 0, r)$ is independent of (k, r) , $\varphi^*(k, 0, r) = A$, say; hence the definition

$$\varphi^*(k, \varepsilon, r) = A \text{ for } k \in \mathbb{R}^2, \varepsilon \leq 0, r \in (\alpha_0, \beta_0)$$

yields the required extension.

To verify (6.3) we use the chain-rule to differentiate $\varphi^*(k, \varepsilon, r) = \hat{\varphi}(h)$, with h given by (6.1), and then apply (4.16), (4.17), and (6.1). $\square$

Proposition 6.1. *There are a compatible neighborhood $\mathcal{K}$ of $\mathcal{E}_0$ and a C^1 function $F = (F_1, F_2)$ on $\mathcal{K}$, with*

$$\nabla F_i = 0 \text{ on } \mathcal{E}_0, \quad (6.4)$$

such that: given $(k, \varepsilon, r) \in \mathcal{K}^+$ and $\Delta = D(k, \varepsilon, r)$,

$$k = F(k, \varepsilon, r) \quad (6.5)$$

if and only if Δ is a solution of $\hat{\mathcal{S}}_{\varepsilon r}$.

Proof. By (3.9) and (5.6),

$$Q_i(\Delta_0) = B_i, \quad (6.6)$$

hence, letting $\xi_1 = \alpha_0$, $\xi_2 = \beta_0$, we conclude from (5.7) and Proposition 4.3 that

$$Q_i(\Delta) = B_i + q_i(\Delta), \quad Q_i(\Delta) z_i(\Delta) = \xi_i B_i + p_i(\Delta) \quad (6.7)$$

with

$$p_i, q_i \in NR(\Omega_0), \quad p_i(\Delta_0) = q_i(\Delta_0) = 0.$$

Note also that

$$\ln h_i = \mu_i k_i - c_i(r)/\varepsilon. \quad (6.8)$$

Thus if we substitute (6.7) into (5.8)₃, appeal to Lemma 6.1, and use (3.10), (3.12), we conclude that there are a compatible neighborhood $\mathcal{K}$ of $\mathcal{E}_0$ and functions

$$S_0, S_1 \in C^1(\mathcal{K}) \text{ with } \nabla S_0 = \nabla S_1 = 0 \text{ on } \mathcal{E}_0$$

such that

$$(\beta_0 - \alpha_0) I_0^*(k, \varepsilon, r) = -k_1 - k_2 + 2(\beta_0 - \alpha_0) \varepsilon^{-1} + S_0(k, \varepsilon, r),$$

$$(\beta_0 - \alpha_0) I_1^*(k, \varepsilon, r) = -\alpha_0 k_1 - \beta_0 k_2 + 2(\beta_0 - \alpha_0) r \varepsilon^{-1} + S_1(k, \varepsilon, r)$$

for all $(k, \varepsilon, r) \in \mathcal{K}^+$. Thus we if define

$$F_1 = (\beta_0 - \alpha_0)^{-1} (\beta_0 S_0 - S_1), \quad F_2 = (\beta_0 - \alpha_0)^{-1} (S_1 - \alpha_0 S_0),$$

then (6.4) holds and the desired conclusion follows. $\square$

We will make use of the following immediate corollary of the implicit function theorem; in that corollary F and $\mathcal{K}$ are as in Proposition 6.1, and we use the fact that, by (6.4),

$$F_i(k, 0, r) = a_i \quad (= \text{constant}) \quad (6.9)$$

for all $k \in \mathbb{R}^2$, $r \in (\alpha_0, \beta_0)$.

Proposition 6.2. *Choose $\delta > 0$ and let $\mathcal{R} = \mathcal{R}_\delta = [\alpha_0 + \delta, \beta_0 - \delta]$. Then there are a neighborhood $\mathcal{B} = \mathcal{B}_\delta$ of $a \in \mathbb{R}^2$, an $\bar{\varepsilon} = \bar{\varepsilon}_\delta > 0$ with $\mathcal{B} \times [-\bar{\varepsilon}, \bar{\varepsilon}] \times \mathcal{R} \subset \mathcal{K}$, and a C^1 function*

$$k_\delta: [-\bar{\varepsilon}, \bar{\varepsilon}] \times \mathcal{R} \rightarrow \mathcal{B}$$

such that: for each $(\varepsilon, r) \in [-\bar{\varepsilon}, \bar{\varepsilon}] \times \mathcal{R}$, (6.5) has exactly one solution $k \in \mathcal{B}$, and

$$k = k_\delta(\varepsilon, r).$$

Let k_δ be as in Proposition 6.2. Then we may conclude from Proposition 6.1 that $\Delta(\varepsilon, r) = D(k_\delta(\varepsilon, r), \varepsilon, r)$ solves $\hat{\mathcal{S}}_{er}$. Further, as k_δ is bounded, (6.1) implies that

$$\pi(\Delta(\varepsilon, r)) = O(e^{-C/\varepsilon}) \quad (6.10)$$

as $\varepsilon \rightarrow 0$ (uniformly for $r \in \mathcal{R}_\delta = [\alpha_0 + \delta, \beta_0 - \delta]$, and, since π is diffeomorphic on $\bar{\Sigma}$, this leads to the first two estimates in (3.4). (Here and in what follows, C and C_1 denote constants > 0 which may vary from line to line.) The remaining estimates follow from (4.18), (6.10), and the fact that, since α_σ and β_σ are smooth in σ near σ_0 ,

$$\alpha_\sigma = \alpha_0 + O(e^{-C/\varepsilon}), \quad \beta_\sigma = \beta_0 + O(e^{-C/\varepsilon}).$$

We will complete the proof by showing that uniqueness of the solution $k = k_\delta(\varepsilon, r)$ to the equation $k = F(k, \varepsilon, r)$ implies uniqueness for the unscaled problem $\hat{\mathcal{S}}_{er}$. This will be facilitated by the notation

$$\text{Sol}(\varepsilon, r) = \{\Delta \in \Sigma: \Delta \text{ is a solution of } \hat{\mathcal{S}}_{er}\}. \quad (6.11)$$

Let $\Delta \in \Sigma$. Then, since $\alpha_\sigma < z_1(\Delta) < \zeta_\sigma < z_2(\Delta) < \beta_\sigma$, it follows from (2.17), (2.18), and (ii) of Proposition 2.1 that $f_\Delta(z) > 0$ for $z_1(\Delta) < z < z_2(\Delta)$. Thus, by (2.20),

$$z_1(\Delta) I_0(\Delta) < I_1(\Delta) < z_2(\Delta) I_0(\Delta);$$

hence (2.21) and (3.2) lead to the conclusion that

$$z_1(\Delta) < r < z_2(\Delta), \quad \Delta \in \text{Sol}(\varepsilon, r). \quad (6.12)$$

Next, by (5.8)₃, on Ω_0

$$I_n + Q_1 z_1^n \ln \pi_1 + Q_2 z_2^n \ln \pi_2 = \varphi_n \in C(\Omega_0).$$

Thus, since $z_i \in C(\Omega_0)$, while $Q_i, z_2 - z_1 \in C^+(\Omega_0)$, we may use (2.21) and (3.2) to conclude that

$$\ln \pi_i(\Delta) = -\varepsilon^{-1} G_i(\Delta, r) + H_i(\Delta) \quad \text{for } \Delta \in \text{Sol}(\varepsilon, r) \cap \Omega_0, \quad (6.13)$$

where

$$G_i \in C(\Omega_0 \times \mathbb{R}), \quad H_i \in C(\Omega_0) \quad (6.14)$$

(are independent of ε) with

$$G_1(\Delta, r) = \frac{2[z_2(\Delta) - r]}{Q_1(\Delta)[z_2(\Delta) - z_1(\Delta)]}, \quad G_2(\Delta, r) = \frac{2[r - z_1(\Delta)]}{Q_2(\Delta)[z_2(\Delta) - z_1(\Delta)]}. \quad (6.15)$$

Choose $\delta > 0$. By (4.18), $z_1(\Delta) \rightarrow \alpha_0$ and $z_2(\Delta) \rightarrow \beta_0$ as $\Delta \rightarrow \Delta_0$ ($\Delta \in \bar{\Sigma}$). We now assume that $\Omega_0 = \Omega_0(\delta)$ is such that

$$|z_i(\Delta) - r| \geq \frac{\delta}{2} \quad \text{for } \Delta \in \Omega_0, r \in \mathcal{R}_\delta.$$

Then by (6.15),

$$G_i(\Delta, r) \geq C \quad \text{for } \Delta \in \Omega_0, r \in \mathcal{R}_\delta,$$

and (6.13), (6.14) yield the estimate

$$\pi_i(\Delta) \leq C_i e^{-C/\varepsilon} \quad \text{for } (\Delta, \varepsilon, r) \in \text{Sol}_\delta, \quad (6.16)$$

where

$$\text{Sol}_\delta = \{(\Delta, \varepsilon, r) : \Delta \in \text{Sol}(\varepsilon, r) \cap \Omega_0, \varepsilon > 0, r \in \mathcal{R}_\delta\}.$$

Further, since the numerator of $G_i(\cdot, r)$ belongs to $NR(\Omega_0)$ uniformly (in the obvious sense) for $r \in \mathcal{R}_\delta$, while the denominator is independent of r and belongs to $NR^+(\Omega_0)$, it follows that $G_i(\cdot, r) \in NR(\Omega_0)$ uniformly for $r \in \mathcal{R}_\delta$. Thus, since

$$G_i(\Delta_0, r) = c_i(r)$$

(cf. (3.10)), if we define

$$g_i(\Delta, r) = G_i(\Delta, r) - c_i(r), \quad (6.17)$$

then

$$|g_i(\Delta, r)| \leq C_1 |\pi(\Delta)|^C,$$

and (6.16) implies

$$\varepsilon^{-1} |g_i(\Delta, r)| \leq C_1 e^{-C/\varepsilon} \quad \text{for } (\Delta, \varepsilon, r) \in \text{Sol}_\delta. \quad (6.18)$$

On the other hand, writing $K_i(\Delta, \varepsilon, r)$ for the components of $K(\Delta, \varepsilon, r)$, we conclude from (6.8), (6.13), and (6.17) that

$$\mu_i K_i(\Delta, \varepsilon, r) = H_i(\Delta) - \varepsilon^{-1} g_i(\Delta, r) \quad \text{for } (\Delta, \varepsilon, r) \in \text{Sol}_\delta,$$

and hence, as $\mu_i > 0$, we may use (6.14) and (6.18) to arrive at

$$|K(\Delta, \varepsilon, r)| \leq C \quad \text{for } (\Delta, \varepsilon, r) \in \text{Sol}_\delta. \quad (6.19)$$

Thus, since $\mathcal{H}$ (of Proposition 6.1) is a neighborhood of $\mathcal{E}_0$ (cf. (6.2)), there is an $\varepsilon_1 > 0$ such that

$$(K(\Delta, \varepsilon, r), \varepsilon, r) \in \mathcal{H}^+ \quad \text{for } (\Delta, \varepsilon, r) \in \text{Sol}_\delta, \varepsilon \in (0, \varepsilon_1).$$

Therefore Proposition 6.1 implies

$$F(K(\Delta, \varepsilon, r), \varepsilon, r) = K(\Delta, \varepsilon, r) \quad \text{for } (\Delta, \varepsilon, r) \in \text{Sol}_\delta, \varepsilon \in (0, \varepsilon_1)$$

and this, with (6.9) and (6.19), yields

$$K(\Delta, \varepsilon, r) \in \mathcal{B} \quad \text{for } (\Delta, \varepsilon, r) \in \text{Sol}_\delta, \quad \varepsilon \in (0, \varepsilon_2) \quad (6.20)$$

for some $\varepsilon_2 \in (0, \varepsilon_1)$, with $\mathcal{B}$ as in Proposition 6.2. In view of that proposition, (6.20) clearly implies the uniqueness assertion of Theorem 3.1. The proof of Theorem 3.1 is now complete. $\square$

Let $z_{\varepsilon r}(t)$ be the simple solution of $\mathcal{S}_{\varepsilon r}$ corresponding to the Maxwell solution $\Delta(\varepsilon, r)$. Then $z_{\varepsilon r}(t)$ is strictly increasing in t . We denote by $T_1(\varepsilon, r)$ the "time" $z_{\varepsilon r}(t)$ takes to go from its initial position $z_{\varepsilon r}(-\varepsilon^{-1})$ to $z_{\varepsilon r}(-\varepsilon^{-1}) + \varepsilon$, by $T_2(\varepsilon, r)$ the time to go from $z_{\varepsilon r}(\varepsilon^{-1}) - \varepsilon$ to $z_{\varepsilon r}(\varepsilon^{-1})$. The following estimates for $T_i(\varepsilon, r)$ will be useful in showing that, as $\varepsilon \rightarrow 0$, $u_{\varepsilon r}(x) = z_{\varepsilon r}(x/\varepsilon)$ approaches the single-interface solution (1.9).

Proposition 6.3. *For each fixed $r \in (\alpha_0, \beta_0)$,*

$$T_i(\varepsilon, r) = \frac{l_i}{\varepsilon} + O(\ln \varepsilon) \quad (6.21)$$

as $\varepsilon \rightarrow 0$, where

$$l_1 = \frac{2(\beta_0 - r)}{\beta_0 - \alpha_0}, \quad l_2 = \frac{2(r - \alpha_0)}{\beta_0 - \alpha_0}. \quad (6.22)$$

Proof. Clearly,

$$T_1(\varepsilon, r) = J_0(\Delta(\varepsilon, r), \varepsilon)$$

with J_0 given by (5.22). In view of (5.15), (6.1), (6.10), and (6.19),

$$\lambda(\Delta(\varepsilon, r)) = O(e^{-C/\varepsilon}), \quad \ln \lambda(\Delta(\varepsilon, r)) = -\frac{c_1(r)}{2\varepsilon} + O(1),$$

the first of which implies

$$\ln [\varepsilon + \lambda(\Delta(\varepsilon, r)) + (\varepsilon^2 + 2\lambda(\Delta(\varepsilon, r))\varepsilon)^{\frac{1}{2}}] = O(\ln \varepsilon).$$

Further, (4.16)₁, (6.6), and (6.10) imply that

$$\mathcal{Q}_1(\Delta(\varepsilon, r)) = B_1 + O(e^{-C/\varepsilon}).$$

Thus, since

$$c_1(r) = \frac{l_1}{B_1}$$

(cf. (3.10)), the above estimates with (5.22) yield (6.21) for $i = 1$. The analysis for $i = 2$ is no different. $\square$

7. Necessary conditions for existence

In view of (4.7) and (6.12), a necessary condition for the existence of a simple solution to problem $\mathcal{S}_{\varepsilon r}$ is that

$$\alpha_0 < r < \beta_0. \quad (7.1)$$

We can actually prove a stronger result.

Theorem 7.1. *Given $\varepsilon > 0$ there is a $\delta_\varepsilon > 0$ such that Problem $\mathcal{S}_{\varepsilon r}$ has no simple solution when $r < \alpha_0 + \delta_\varepsilon$, and none when $r > \beta_0 - \delta_\varepsilon$.*

Proof. Assume the theorem is false. Then, by (7.1), there exist sequences $\{\Delta_N\}$, $\{r_N\}$ such that

$$\Delta_N \in \text{Sol}(\varepsilon, r_N), \quad r_N \rightarrow \alpha_0 \text{ or } r_N \rightarrow \beta_0.$$

Assume that

$$r_N \rightarrow \beta_0.$$

Since $\Delta_N \in \Sigma$, we may suppose (perhaps by extracting a subsequence) that

$$\Delta_N \rightarrow \tilde{\Delta} \in \bar{\Sigma}.$$

By (4.7) and (6.12), $r_N < z_2(\Delta_N) < \beta_0$; hence

$$z_2(\tilde{\Delta}) = \beta_0.$$

Thus $\tilde{\Delta} \notin \partial_3 \Sigma$, since $z_2(\Delta) = \zeta_\sigma$ on $\partial_3 \Sigma$. On the other hand, since (2.21)₁ yields

$$\lim_{N \rightarrow \infty} I_0(\Delta_N) = 2\varepsilon^{-1} < \infty$$

(ε is fixed), it follows from (5.21) that $\tilde{\Delta} \notin \partial_1 \Sigma \cup \partial_2 \Sigma$. Thus $\tilde{\Delta} \in \Sigma$, and if we take $\Delta = \Delta_N$ and $r = r_N$ in (2.21) and pass to the limit as $N \rightarrow \infty$, we find that

$$\tilde{\Delta} \in \text{Sol}(\varepsilon, \beta_0),$$

which contradicts the necessary condition (7.1). A similar contradiction arises when $r_N \rightarrow \alpha_0$. $\square$

Remark. Theorem 3.1 asserts the existence of the Maxwell solution $\Delta(\varepsilon, r)$ for $0 < \varepsilon < \varepsilon_\delta$ and $\alpha_0 + \delta \leq r \leq \beta_0 - \delta$; Theorem 7.1 shows that

$$\varepsilon_\delta \rightarrow 0 \quad \text{as } \delta \rightarrow 0.$$

In the previous section we established the existence and uniqueness of simple solutions of $\mathcal{S}_{\varepsilon r}$ with Δ near the Maxwell point Δ_0 . There are, in fact, simple solutions of $\mathcal{S}_{\varepsilon r}$ with $|\Delta - \Delta_0|$ not small, and their existence may be established by finding an appropriate scaling. The issue of existence, however, is not crucial; as we shall see, such solutions have higher energy than those close to Δ_0 . Our proof of this assertion requires locating where such possible solutions may exist, a topic which we now address. In particular, we now show that solutions Δ of $\mathcal{S}_{\varepsilon r}$ which are not Maxwell have σ bounded away from σ_0 , have Δ close to $\partial \Sigma$, and have corresponding $z(t)$ close to the constant solution

$$z(t) \equiv r, \quad W'(r) = \sigma.$$

For convenience, we continue to use the notation

$$\mathcal{R}_\delta = [\alpha_0 + \delta, \beta_0 - \delta].$$

Theorem 7.2. Choose $\varrho, \delta > 0$. Then there exist an $\varepsilon_0 = \varepsilon_0(\delta, \varrho) > 0$ and a $\varrho_0 = \varrho_0(\delta) > 0$ such that: for

$$\Delta \in \text{Sol}(\varepsilon, r), \quad \varepsilon \in (0, \varepsilon_0), \quad r \in \mathcal{R}_\delta,$$

Δ not the Maxwell solution for (ε, r) ,

it follows that:

(i) σ is bounded away from σ_0 :

$$|\sigma - \sigma_0| > \varrho_0;$$

(ii) Δ lies close to $\partial_1 \Sigma \cup \partial_2 \Sigma$:

$$\pi_2(\Delta) < \varrho \text{ for } \sigma < \sigma_0, \quad \pi_1(\Delta) < \varrho \text{ for } \sigma > \sigma_0;$$

(iii) $|W'(r) - \sigma| < \varrho$, while

$$|r - \beta_\sigma| < \varrho \text{ for } \sigma < \sigma_0, \quad |r - \alpha_\sigma| < \varrho \text{ for } \sigma > \sigma_0.$$

Proof. Let

$$d(\Delta) = \text{distance between } \Delta \text{ and } \partial_1 \Sigma \cup \partial_2 \Sigma,$$

$$\mathcal{D}_\lambda(\mathcal{T}) = \{(\sigma, b) \in \bar{\Sigma}: |\sigma - \lambda| \leq \mathcal{T}\},$$

$$\mathcal{B}(\mathcal{T}) = \{\Delta \in \mathcal{D}_{\sigma_0}(\mathcal{T}): d(\Delta) \leq \mathcal{T}\}.$$

Take $\varrho_0 = \varrho_0(\delta)$, $\varepsilon_1 = \varepsilon_1(\delta)$ such that for all $\varepsilon \in (0, \varepsilon_1)$ and $r \in \mathcal{R}_\delta$, the Maxwell solution $\Delta(\varepsilon, r)$ is the only solution of $\mathcal{P}_{\varepsilon r}$ in $\mathcal{B}(\varrho_0)$.

Choose $\varrho > 0$. In view of (4.1), (4.5), (4.6), (4.8)₃, and (4.15)_{2,3},

$$z_i(\underline{\Delta}) = \underline{\beta}, \quad z_i(\bar{\Delta}) = \bar{\alpha},$$

$$\pi_2(\underline{\Delta}) = 0, \quad \pi_1(\bar{\Delta}) = 0.$$

Thus, since $z_i(\Delta)$ and $\pi_i(\Delta)$ are continuous on $\bar{\Sigma}$, while $W'(z)$ is continuous on $(0, \infty)$, we conclude from (2.12) and (6.12) that there is a $\varrho_1 = \varrho_1(\varrho)$ such that (ii) and (iii) hold whenever

$$\Delta \in \mathcal{D}_{\underline{\sigma}}(\varrho) \cup \mathcal{D}_{\bar{\sigma}}(\varrho), \quad \Delta \in \text{Sol}(\varepsilon, r).$$

We assume further that ϱ_0 and ϱ_1 have been chosen such that $\mathcal{B}(\varrho_0)$, $\mathcal{D}_{\underline{\sigma}}(\varrho_1)$, and $\mathcal{D}_{\bar{\sigma}}(\varrho_1)$ are mutually disjoint.

Next, there exist closed subregions Ω_1, Ω_2 of $\bar{\Sigma}$ with the following properties (Figure 5):

(iv) Ω_1 and Ω_2 are consistent with (5.1);

(v) the points of $\bar{\Sigma}$ exterior to

$$\Delta = \Omega_1 \cup \Omega_2 \cup \mathcal{B}(\varrho_0) \cup \mathcal{D}_{\underline{\sigma}}(\varrho_1) \cup \mathcal{D}_{\bar{\sigma}}(\varrho_1)$$

are uniformly bounded away from $\partial_1 \Sigma \cup \partial_2 \Sigma$;

(vi) $\Omega_1, \Omega_2, \mathcal{D}_{\sigma_0}(\varrho_0)$, $\mathcal{D}_{\underline{\sigma}}(\varrho_1)$, and $\mathcal{D}_{\bar{\sigma}}(\varrho_1)$ have disjoint interiors.

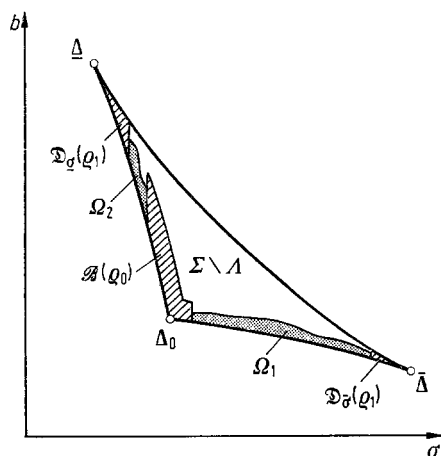


Fig. 5. The sets Ω_1 , Ω_2 , $\mathcal{B}(\varrho_0)$, $\mathcal{D}_\sigma(\varrho_1)$, $\mathcal{D}_\sigma(\varrho_1)$

As is known,¹ $I_0(\Delta)$ is continuous up to $\partial_3 \Sigma \setminus \{\underline{\Delta}, \overline{\Delta}\}$. Thus and by (v), $I_0(\Delta)$ is continuous and hence bounded on the closure of $\Sigma \setminus \Delta$. Consequently, by (2.21)₁, there exists an $\varepsilon_2 \in (0, \varepsilon_1)$ such that for $\varepsilon \in (0, \varepsilon_2)$,

$$\Delta \in \text{Sol}(\varepsilon, r) \quad \text{implies} \quad \Delta \in \Delta.$$

Let $\Delta \in \Omega_1 \cup \Omega_2$. Then, trivially, (i) holds. To complete the proof we have to find an $\varepsilon_0 > 0$ such that (ii) and (iii) are satisfied when, in addition, $\Delta \in \text{Sol}(\varepsilon, r)$ with $\varepsilon \in (0, \varepsilon_0)$.

By (5.8)₁,

$$I_n + Q_1 z_1^n \ln \pi_1 \text{ is continuous on } \Omega_1. \quad (7.2)$$

Thus, since by (2.21)

$$r - z_1(\Delta) = \frac{\varepsilon}{2} [\varphi_1(\Delta) - z_1(\Delta) \varphi_0(\Delta)]$$

for $\Delta \in \text{Sol}(\varepsilon, r) \cap \Omega_1$,

$$|r - z_1(\Delta)| \leq C\varepsilon$$

for all such (Δ, ε, r) . Also, in view of (2.21)₁, (5.7), and (7.2) with $n = 0$, for such (Δ, ε, r)

$$\pi_1(\Delta) \leq C_1 e^{-C/\varepsilon}.$$

By (4.18)₁ this implies

$$|z_1(\Delta) - \alpha_\sigma| \leq C_1 e^{-C/\varepsilon}$$

and thus

$$|r - \alpha_\sigma| \leq C\varepsilon.$$

Analogous estimates hold on Ω_2 , and these with the above obviously yield the existence of an ε_0 with the desired properties. $\square$

¹ Cf., e.g., HALE [1969], p. 179; ARNOLD [1978], Ch. 2, § 4.

8. The Maxwell solution has least energy

In view of the variable change $x = \varepsilon t$, $z(t) = u(\varepsilon t)$, the energy (2.1) has the form

$$E_\varepsilon(z) = \varepsilon \int_{-\varepsilon^{-1}}^{\varepsilon^{-1}} [W(z(t)) + \dot{z}(t)^2] dt, \quad (8.1)$$

and we are seeking a minimum over the class

$$\mathcal{Z}_{\varepsilon r} = \left\{ z \in H^1(-\varepsilon^{-1}, \varepsilon^{-1}) : z > 0, \int_{-\varepsilon^{-1}}^{\varepsilon^{-1}} z(t) dt = 2r\varepsilon^{-1} \right\}. \quad (8.2)$$

(A global minimizer $z(t)$ over $\mathcal{Z}_{\varepsilon r}$ yields a global minimizer $u(x) = z(x/\varepsilon)$ of our original problem $\mathcal{P}_{\varepsilon r}$ of Section 2.)

Given any simple solution $z(t)$ of $\mathcal{S}_{\varepsilon r}$, the *reversal*

$$z^R(t) = z(-t)$$

is also a solution of $\mathcal{S}_{\varepsilon r}$, and

$$E_\varepsilon(z^R) = E_\varepsilon(z). \quad (8.3)$$

Thus there are actually two solutions of $\mathcal{S}_{\varepsilon r}$ corresponding to the Maxwell solution $\Delta(\varepsilon, r)$: the associated simple solution $z_{\varepsilon r}(t)$ and the reversal $z_{\varepsilon r}^R(t)$. We write

$$E_{\varepsilon r}^0 = E_\varepsilon(z_{\varepsilon r}) = E_\varepsilon(z_{\varepsilon r}^R)$$

and call $E_{\varepsilon r}^0$ the *energy of the Maxwell solution*.

If we restrict our attention to simple solutions, then we can rewrite the functional $E_\varepsilon(z)$ (restricted to $\mathcal{Z}_{\varepsilon r}$) as a function $E_{\varepsilon r}(\Delta)$ of the admissible pair corresponding to $z(t)$. Indeed, by (2.7), (2.16)₁, and (2.17),

$$W(z) + \dot{z}^2 = 2f_\Delta(z) + \sigma z + b,$$

and hence, using (2.16) to change the variable of integration, we conclude, with the aid of the constraint integral (2.5)₃, that

$$E_{\varepsilon r}(\Delta) = 2(\sigma r + b) + 2\varepsilon \int_{z_1(\Delta)}^{z_2(\Delta)} f_\Delta(z)^{\frac{1}{2}} dz \quad (8.4)$$

for $\Delta \in \text{Sol}(\varepsilon, r)$. Of course,

$$E_{\varepsilon r}^0 = E_{\varepsilon r}(\Delta(\varepsilon, r)).$$

Remark. The critical points of $E_{\varepsilon r}(\Delta)$ ($\Delta \in \Sigma$) are exactly the solutions of (2.21); indeed, a simple calculation using (2.7) and (4.1) shows that

$$\frac{\partial}{\partial \sigma} E_{\varepsilon r}(\Delta) = 2r - \varepsilon I_1(\Delta), \quad \frac{\partial}{\partial b} E_{\varepsilon r}(\Delta) = 2 - \varepsilon I_0(\Delta).$$

We now use (8.4) to obtain an estimate for $E_{\varepsilon r}^0$. With this in mind, we let S denote the constant

$$S = 2 \int_{\alpha_0}^{\beta_0} [W(z) - W(\beta_0) - \sigma_0(z - \beta_0)]^{\frac{1}{2}} dz. \quad (8.5)$$

Theorem 8.1. *The energy of the Maxwell solution has the asymptotic form*

$$E_{er}^0 = 2(\sigma_0 r + b_0) + \varepsilon S + O(e^{-C/\varepsilon}) \quad (8.6)$$

as $\varepsilon \rightarrow 0$, uniformly for r in any closed subinterval of (α_0, β_0) .

Proof. Write $\varphi(\Delta)$ for the integral in (8.4). Then, by Lemma 4.2, $\hat{\varphi}(h)$ is nearly regular at $h = 0$, and the desired estimate follows from (3.1), (3.4), (4.16)₁, and (6.10). $\square$

To within a term of $O(e^{-C/\varepsilon})$, εS is the *interfacial energy* of the Maxwell solution; indeed, $\varepsilon S + O(e^{-C/\varepsilon})$ is the difference between the actual energy E_{er}^0 and the energy $2(\sigma_0 r + b_0)$ of the global minimizer when $\varepsilon = 0$. Interestingly, the term εS is independent of r ; it represents the interfacial energy, as deduced by CAHN & HILLIARD [1958], for a medium of infinite extent.

We now show that for small ε the Maxwell solution minimizes energy. Since $\Delta(\varepsilon, r)$ does not exist uniformly in r (cf. the Remark following Theorem 7.1), we can at most expect this minimization to occur for parameter ranges of the form $\alpha_0 + \delta \leq r \leq \beta_0 - \delta$, $0 < \varepsilon < \varepsilon_\delta$. With this in mind, we let

$$\mathcal{R}_\delta = [\alpha_0 + \delta, \beta_0 - \delta],$$

and we introduce the following definition in which $\mathcal{L}_{er}$, for each (ε, r) , is a subclass of the family $\mathcal{Z}_{er}$ (cf. (8.2)) over which we are seeking a minimum. We say that the **Maxwell solution for (ε, r) minimizes energy over $\mathcal{L}_{er}$ —for r in closed subsets of (α_0, β_0) and small ε** —if given any $\delta > 0$ there is an $\varepsilon_\delta > 0$ such that for $r \in \mathcal{R}_\delta$ and $\varepsilon \in (0, \varepsilon_\delta)$,

$$E_{er}^0 < E_\varepsilon(z) \quad \text{for every } z \in \mathcal{L}_{er}, \quad z \neq z_{er}, z_{er}^R.$$

We will use the phrase “globally minimizes” in place of “minimizes” for the special case in which $\mathcal{L}_{er} = \mathcal{Z}_{er}$.

Proposition 8.1. *For r in closed subsets of (α_0, β_0) and small ε , the Maxwell solution for (ε, r) minimizes energy over:*

- (i) *the class of all constant solutions of $\mathcal{S}_{er}$;*
- (ii) *the class of all simple solutions of $\mathcal{S}_{er}$.*

Proof. Let z be a constant solution of $\mathcal{S}_{er}$. The constraint (2.5) then implies that

$$z(t) \equiv r,$$

and by (8.1) the corresponding energy is

$$E_\varepsilon(r) = 2W(r).$$

Thus, in view of (2.7), (3.1), and (8.6),

$$E_\varepsilon(r) - E_{er}^0 = 2\{\Phi_{\sigma_0}(r) - \Phi_{\sigma_0}(\beta_0)\} + O(\varepsilon),$$

uniformly for r in any $\mathcal{R}_\delta$, and conclusion (i) follows from the fact that $\Phi_{\sigma_0}(z)$ has a strict global minimum at $z = \alpha_0, \beta_0$ and r is bounded away from α_0 and β_0 .

To establish (ii) it suffices to show that: (I) given any $\delta > 0$ there is an $\varepsilon_\delta > 0$ such that for $r \in \mathcal{R}_\delta$ and $\varepsilon \in (0, \varepsilon_\delta)$,

$$E_{er}^0 < E_{er}(\Delta)$$

for $\Delta \in \text{Sol}(\varepsilon, r)$, $\Delta = (\sigma, b) \neq \Delta(\varepsilon, r)$, and $\sigma \leq \sigma_0$; (II) the same conclusion applies for $\sigma > \sigma_0$. We will consider only the case $\sigma \leq \sigma_0$; the proof for $\sigma > \sigma_0$ is completely analogous.

Choose $\delta > 0$. By (8.4), (8.6), and (3.8)₂, for $\Delta \in \text{Sol}(\varepsilon, r)$ and $r \in \mathcal{R}_\delta = [\alpha_0 + \delta, \beta_0 - \delta]$,

$$E_{er}(\Delta) - E_{er}^0 = R(\sigma, r) + \pi_2(\Delta) + O(\varepsilon), \quad (8.7)$$

where

$$R(\sigma, r) = 2[(\sigma - \sigma_0)r + \Phi_\sigma(\beta_\sigma) - \Phi_{\sigma_0}(\beta_0)],$$

and where the $O(\varepsilon)$ term is uniform for $\Delta \in \bar{\Sigma}$, $r \in \mathcal{R}_\delta$.

In view of the form of (8.7), it suffices to find constants $\varepsilon_0, C_0 > 0$ such that

$$R(\sigma, r) + \pi_2(\Delta) \geq C_0 \quad (8.8)$$

for

$$r \in \mathcal{R}_\delta, \quad \varepsilon \in (0, \varepsilon_0), \quad \Delta \in \text{Sol}(\varepsilon, r), \quad \Delta \neq \Delta(\varepsilon, r), \quad \sigma \leq \sigma_0.$$

For any solution, $r < \beta_\sigma$ (cf. (6.12) and Proposition 2.2), so we need only study $R(\sigma, r)$ on the set

$$\Gamma = \{(\sigma, r): \alpha_0 \leq r \leq \beta_\sigma, \underline{\sigma} \leq \sigma \leq \sigma_0\}.$$

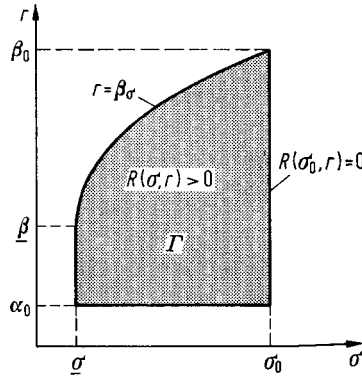


Fig. 6. The set Γ

By Proposition 4.1, β_σ is strictly increasing in σ for $\sigma \leq \sigma \leq \sigma_0$; hence Γ has the form shown in Figure 6. Moreover, by (2.13) and (4.2), for $(\sigma, r) \in \Gamma$,

$$\frac{\partial}{\partial \sigma} R(\sigma, r) = r - \beta_\sigma, \quad R(\sigma_0, r) = 0,$$

so that

$$R(\sigma, r) > 0 \text{ for } \sigma < \sigma_0.$$

Thus, if $\varrho_0 > 0$ is as in Theorem 7.2, there are constants $C_0, \eta > 0$ such that

$$R(\sigma, r) \geq 2C_0 \quad \text{for } (\sigma, r) \in \Gamma, \quad r \leq \beta_\sigma - \eta, \quad \sigma \leq \sigma_0 - \varrho_0. \quad (8.9)$$

Choose $\varrho > 0$ such that $\varrho < C_0, \eta$. Then Theorem 7.2 implies the existence of an $\varepsilon_0 > 0$ such that $\varepsilon \in (0, \varepsilon_0)$, $r \in \mathcal{R}_\delta$, $\Delta \in \text{Sol}(\varepsilon, r)$, $\Delta \neq \Delta(\varepsilon, r)$, $\sigma \leq \sigma_0$

together imply

$$\sigma < \sigma_0 - \varrho_0, \quad |r - \beta_\sigma| < \eta, \quad \pi_2(\Delta) < C_0,$$

which, with (8.9), implies (8.8). $\square$

Solutions of $\mathcal{S}_{er}$ must be either: constant, simple, reversals of simple solutions, or nonmonotonic. By (8.3) and Proposition 8.1, the Maxwell solution has least energy when compared to the first three types of solutions. We now turn our attention to nonmonotonic solutions.

We say that $z \in \mathcal{Z}_{er}$ is a **local minimizer** if for some neighborhood $\mathcal{N}$ of z in $\mathcal{Z}_{er}$,

$$E(z) \leq E(y) \quad \text{for all } y \in \mathcal{N}. \quad (8.10)$$

(The topology on $\mathcal{Z}_{er}$ is that induced by the norm on $H^1(-\varepsilon^{-1}, \varepsilon^{-1})$.)

Theorem 8.2.¹ *Nonmonotonic solutions of $\mathcal{S}_{er}$ cannot be even local minimizers.*

Proof. For convenience, let $l = \varepsilon^{-1}$. Let z be a nonmonotonic solution. We will show that z cannot be a local minimizer by showing that² the second variation

$$\frac{d^2}{d\xi^2} E_\varepsilon(z + \xi\eta)|_{\xi=0} \quad (8.11)$$

is strictly negative at some $\eta \in H^1(-l, l)$ with

$$\int_{-l}^l \eta(t) dt = 0. \quad (8.12)$$

(This latter restriction insures that $z + \xi\eta \in \mathcal{Z}_{er}$ for all sufficiently small ξ .) Write $\varepsilon J(z, \eta)$ for the second variation (8.11); then (8.1) yields

$$J(z, \eta) = \int_{-l}^l [2\dot{\eta}^2 + W''(z)\eta^2] dt.$$

Since nonconstant solutions of $\mathcal{S}_{er}$ lie on closed orbits, and since z is nonmonotonic, z must repeat: there is a $T \in (-l, l)$ such that

$$z(T) = z(-l), \quad \dot{z}(T) = 0 \quad (= \dot{z}(-l)). \quad (8.13)$$

Let

$$\eta_0(t) = \begin{cases} \dot{z}(t), & -l \leq t \leq T \\ 0, & T \leq t \leq l, \end{cases}$$

let $\eta_1(t)$ be any $H^1(-l, l)$ function that satisfies (8.12) and

$$\eta_1(-l) = 1, \quad \eta_1(t) = 0 \quad \text{for } T \leq t \leq l, \quad (8.14)$$

and let

$$\eta(t) = \eta_0(t) + \kappa\eta_1(t)$$

¹ For W a quartic, NOVICK-COHEN [1984] has obtained this result in certain limiting cases. Our theorem was arrived at independently.

² Cf., e.g., the discussion given by CARR, GURTIN & SLEMROD [1983], § 3.

with $\varkappa$ to be determined. Then $\eta \in H^1(-l, l)$ and satisfies (8.12), and

$$J(z, \eta) = \int_{-l}^T [2\dot{z}^2 + W''(z) \dot{z}^2] dt + 2\varkappa \int_{-l}^T [2\dot{z}\dot{\eta}_1 + W''(z) \dot{z}\eta_1] dt + O(\varkappa^2). \quad (8.15)$$

Further, by (2.3),

$$2\ddot{z} = W''(z) \dot{z},$$

and if we replace $W''(z) \dot{z}$ by $\ddot{z}$ in (8.15), we find, using (8.13) and (8.14), that

$$J(z, \eta) = -4\varkappa \ddot{z}(-l) + O(\varkappa^2). \quad (8.16)$$

Since z is not constant, $\ddot{z}(-l) \neq 0$; it is therefore clear from (8.16) that we can choose $\varkappa$ such that $J(z, \eta) < 0$. $\square$

By (8.3), Proposition 8.1, and Theorem 8.2, the Maxwell solution has least energy when compared to all other solutions of $\mathcal{S}_{er}$. Thus, since minimizers must necessarily be solutions of $\mathcal{S}_{er}$, we have

Theorem 8.3. *The Maxwell solution for (ε, r) globally minimizes energy for r in closed subsets of (α_0, β_0) and small ε .*

9. Results for Problem $\mathcal{P}_{er}$

We now recast our results within the framework of the original problem $\mathcal{P}_{er}$, as stated in Section 2. The Euler-Lagrange equation, natural boundary condition, and constraint take the form

$$\begin{aligned} 2\varepsilon^2 u'' &= W'(u) - \sigma, \\ u'(\pm 1) &= 0, \\ \int_{-1}^1 u(x) dx &= 2r. \end{aligned} \quad (9.1)$$

For $r \leq \alpha_0$ or $r \geq \beta_0$, the constant

$$u(x) \equiv r,$$

with $\sigma = W'(r)$, is the only solution of the system (9.1) and hence is the global minimizer for Problem $\mathcal{P}_{er}$.

For $r \in (\alpha_0, \beta_0)$, (9.1) has solutions

$$\begin{aligned} u_{er}(x) &= z_{er}(x/\varepsilon), \\ u_{er}^R(x) &= u_{er}(-x), \end{aligned}$$

with $z_{er}(t)$ the simple solution corresponding to the Maxwell solution $\Delta(\varepsilon, r)$. Writing $\Delta(\varepsilon, r) = (\sigma_{er}, b_{er})$, we see that

$$\sigma_{er} = W'(u_{er}) - 2\varepsilon^2 u_{er}''.$$

The solutions $u_{er}(x)$ and $u_{er}^R(x)$ have the following properties:

Theorem 9.1.

(i) $u_{\varepsilon r}(x)$ and $u_{\varepsilon r}^R(x)$ are the global minimizers of Problem $\mathcal{P}_{\varepsilon r}$ for r in closed subsets of (α_0, β_0) and small ε .

(ii) For each $r \in (\alpha_0, \beta_0)$ and $x \in [-1, 1]$,

$$u_{\varepsilon r}(x) \rightarrow u_0(x) \quad (x \neq -1 + l_1),$$

$$u_{\varepsilon r}^R(x) \rightarrow u_0(-x) \quad (x \neq -1 + l_2)$$

as $\varepsilon \rightarrow 0$, where $u_0(x)$ is the single-interface solution (1.9) of the problem with $\varepsilon = 0$,

(iii) $\sigma_{\varepsilon r} = \sigma_0 + O(e^{-C/\varepsilon})$ as $\varepsilon \rightarrow 0$, uniformly for r in closed subsets of (α_0, β_0) .

Proof. Condition (i) is an obvious consequence of Theorem 8.3, while (iii) follows from (3.4)₁. To verify (ii), note that, since $u_{\varepsilon r}(x) = z_{\varepsilon r}(x/\varepsilon)$, Proposition 6.3 and the definition of $T_1(\varepsilon, r)$ yield

$$|z_1(\Delta(\varepsilon, r)) - u_{\varepsilon r}(x)| \leq \varepsilon \quad \text{for } -1 \leq x \leq -1 + l_1 + O(\varepsilon \ln \varepsilon),$$

$$|z_2(\Delta(\varepsilon, r)) - u_{\varepsilon r}(x)| \leq \varepsilon \quad \text{for } 1 - l_2 + O(\varepsilon \ln \varepsilon) \leq x \leq 1.$$

Since $l_1 + l_2 = 2$, (ii) follows from this estimate and (3.4)_{2,3}. $\square$

Note also that $u_{\varepsilon r}(x)$ is strictly increasing in x , while $u_{\varepsilon r}^R(x)$ is strictly decreasing. In addition, we have the following estimate for the L^2 -norm of $u'_{\varepsilon r}$:

$$\|u'_{\varepsilon r}\|_{L^2(-1,1)} = \left(\frac{S}{2\varepsilon}\right)^{\frac{1}{2}} + O(e^{-C/\varepsilon}) \rightarrow \infty \text{ as } \varepsilon \rightarrow 0,$$

with S the constant (8.5). This last result is a consequence of the calculation (cf. Section 8)

$$\int_{-1}^1 (u')^2 dx = \varepsilon^{-1} \int_{-\varepsilon^{-1}}^{\varepsilon^{-1}} z^2 dt = \varepsilon^{-1} \int_{z_1(\Delta)}^{z_2(\Delta)} f_{\Delta}(x)^{\frac{1}{2}} dz = \frac{S}{2\varepsilon} + O(e^{-C/\varepsilon}),$$

where we have written $u = u_{\varepsilon r}$, $z = z_{\varepsilon r}$, $\Delta = \Delta(\varepsilon, r)$.

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